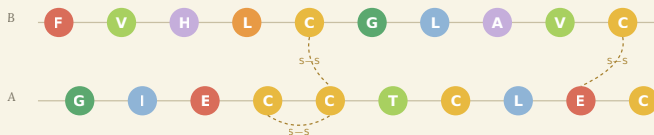


VOLUME ONE • SEVENTY THERAPEUTICS

The Peptide Pocket Guide

*A reference to seventy peptide therapeutics,
biologics, and mixtures.*

INSULIN • THE FIRST PEPTIDE DRUG

**FADY HANNAH-SHMOUNI, MD FRCPC**

Clinical Endocrinologist & Geneticist
University of British Columbia
Founder & CEO, Hormonally.ai

2026 EDITION

The Peptide Pocket Guide

Seventy entries: peptide therapeutics, peptide-derived biologics, and peptide mixtures

A bench-side reference covering the major regulated and unregulated peptides — with GRADE evidence ratings, mechanism, dosing, and regulatory status, one page per therapeutic.

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How to use this guide

This pocket guide covers seventy entries — peptide therapeutics, peptide-derived biologics, peptide mixtures, and one non-peptide comparator (orforglipron, included because it competes directly with established peptide GLP-1 RAs in the same clinical decision). FDA- and EMA-approved drugs sit alongside investigational and compounding-accessible agents commonly encountered in modern practice.

Each entry occupies **one page** and follows a consistent template: the header carries the GRADE rating and therapeutic class; then name and brand names, mechanism of action, indications, dosing, evidence summary, safety, adverse reactions, a clinical caveat, and up to three key references. The body type is one uniform size throughout the book, and wording is kept concise so every peptide fits a single page without a mechanism diagram or amino-acid sequence taking up space.

READING AN ENTRY

The header carries two things:

- **GRADE letter** — A through D, the evidence rating for the listed clinical use (see the GRADE page for definitions). It appears as a coloured pill at the top-left of every entry.
- **Therapeutic class** — the drug class or mechanism family, at the top-right.

Regulatory and trial status (approved, investigational, compounding-accessible, research-only) is stated in plain language in the status line and clinical caveat, rather than as a separate header flag. **Regulatory and trial statuses reflect information available at publication (2026 edition)** and should be re-verified before clinical or legal reliance.

HOW THE BOOK IS ORGANIZED

Entries are grouped by therapeutic class — metabolic, growth-hormone axis, regenerative, dermatology, bone, somatostatin, melanocortin, reproductive, mitochondrial / longevity, neurological, oncology, and gut / antimicrobial — ordered most-evidence-rich first within each class. The back matter contains clinical pearls, a lab and monitoring reference, an adverse-effects and GH-excess primer, the regulatory landscape (including the Lilly v. FDA retatrutide ruling), and a fully claim-mapped bibliography.

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GRADE evidence ratings

Every entry carries a GRADE letter rating derived from the Hormonal.ai evidence database, which adapts the standard GRADE framework to peptide pharmacology:

GRADE A

High-quality evidence. Multiple pivotal RCTs with consistent findings; broadly accepted standard of care for the listed indications.

GRADE B

Moderate-quality evidence. Some RCTs and substantial observational data; conclusions reasonably stable but not yet definitive.

GRADE C

Low-quality evidence. Limited human data, mostly observational or small trials; mechanistic and animal data may be strong but human translation incomplete.

GRADE D

Very low-quality evidence. Mostly preclinical, anecdotal, or industry-driven claims; clinical use carries meaningful uncertainty.

A GRADE C or D rating does not mean a peptide is ineffective — only that the evidence supporting its use does not yet meet the standard of approved therapeutics.

Editorial Standards & Evidence Philosophy

This guide is a clinical reference, not a recommendation or endorsement.

Inclusion does not imply that a peptide is approved, recommended for practice, appropriate for any individual patient, or standard-of-care.

EVIDENCE-FIRST, BY GRADE

Every entry is rated A–D by GRADE methodology, which reflects the strength of *clinical evidence* — not molecular plausibility, popularity, compounding availability, or fashion. An FDA-approved drug with one good RCT may rate lower than an investigational compound with a compelling mechanism; both are contextualized accurately. GRADE A does not mean "use this"; GRADE D does not mean "do not." Clinical decisions require GRADE *plus* regulatory status, contraindications, patient preference, and judgment.

WHY INVESTIGATIONAL AND GRAY-MARKET COMPOUNDS ARE INCLUDED

The guide spans approved drugs, investigational peptides in active trials, 503A-compounded agents, and research-only substances seen in gray-market settings. Inclusion is not endorsement — it reflects clinical reality: agents like BPC-157, TB-500, and Dihexa are used off-label without approval, and clinicians need their preclinical evidence, known harms, and evidentiary gaps to manage patients who use them.

FOUR TIERS, KEPT SEPARATE

Each entry separates what is known from what is hoped, in plain language:

- **Established evidence** — FDA approval, positive Phase 3 RCTs, meta-analysis consensus.
- **Emerging evidence** — Phase 2 human data, small observational studies, mechanism validated in humans.
- **Clinical culture** — usage in anti-aging clinics, bodybuilding, and biohacker communities; documented, never conflated with evidence.
- **Mechanistic plausibility** — in-vitro activity and animal models; not clinical evidence until translated to human efficacy and safety.

A peptide supported by **animal or preclinical data only** has no human efficacy evidence, regardless of GRADE rating or mechanistic appeal.

CONTEXT WITHOUT ENDORSEMENT

Descriptions like "widely used in anti-aging clinics" or "common microdosing protocol" are observations, not endorsements. For Grade C/D, "Research only," "Not approved," or "Gray market" entries, the clinical caveat states plainly whether *use carries unjustifiable risk* or *no human efficacy trials support this use* — factual statements, not hedges. A clinician meeting a patient on an off-label peptide should leave each entry knowing what the culture believes, what the evidence shows, and where the gaps are.

Peptide Fundamentals

Definitions, classification, manufacturing, regulatory grades, and how to read a Certificate of Analysis — the foundation behind every entry in this guide.

WHAT IS A PEPTIDE?

A peptide is a short chain of amino acids linked by peptide bonds. The biochemical distinction from a protein is conventionally one of length: peptides are typically **2–50 residues**; proteins are longer, fold into stable tertiary structures, and usually exceed 10 kDa. The boundary is genuinely fuzzy — insulin (51 residues, two disulfide-linked chains) crosses the threshold by one residue and is variably classified as a polypeptide or a small protein. We list it as a 51-residue exception under the polypeptide class.

THE FDA'S REGULATORY LINE

The biochemical definition above is descriptive; the FDA's definition is operational and consequential. Under the BPCIA (2009) as amended by the Further Consolidated Appropriations Act (2020), the agency draws the peptide/protein line at **40 alpha amino acids**:

- **≤ 40 alpha amino acids** — peptide, regulated as a drug under the FD&C Act (NDA pathway).
- **41–99 alpha amino acids, chemically synthesized** — "chemically synthesized polypeptide", still regulated as a drug under the FD&C Act.
- **> 40 alpha amino acids, not chemically synthesized** (recombinant or extracted) — protein, regulated as a biological product under the PHS Act (BLA pathway).
- **≥ 100 alpha amino acids** — always a protein/biologic regardless of synthesis route.

The classification matters: biologics face additional statutory protections (12-year exclusivity, biosimilar pathway), more rigorous CMC review, and are largely ineligible for 503A pharmacy compounding. Drugs follow the smaller-molecule NDA pathway with shorter exclusivity and a more permissive compounding regime.

Classification & chain length

CLASSIFICATION BY CHAIN LENGTH

CLASS	RESIDUES	EXAMPLES IN THIS GUIDE
Dipeptides & tripeptides	2–3	GHK (3), KPV (3)
Oligopeptides	4–20	Oxytocin (9), GnRH (10), Octreotide (8), Kisspeptin-10 (10), BPC-157 (15)
Polypeptides	20–50	GLP-1 analogues (30–31), Glucagon (29), Calcitonin (32), Tesamorelin (44)
Small proteins	≥ 51	Insulin (51, boundary case), GH (191), Follistatin (315), Cerebrolysin (mixture)

WHY CHAIN LENGTH MATTERS

Residue count drives pharmacology and, near 40 residues, regulatory status:

LENGTH RANGE	PHARMACOLOGY, SYNTHESIS, AND REGULATION
≤ 10 residues	Solid-phase synthesis. Half-life often minutes (oxytocin ≈ 3 min); rapidly cleared by peptidases, needing frequent or sustained-release dosing.
10–30 residues	Solid-phase synthesis still cost-effective. Half-life extended by D-amino-acid substitution (octreotide), N-methylation, cyclization, or PEGylation.
> 30 residues	Recombinant production becomes attractive. Half-life extended by fatty-acid acylation (semaglutide $t_{1/2}$ ~7 days), Fc-fusion (dulaglutide), or depot (lanreotide).

A NOTE ON BIOREGULATORS

One *functional* class cuts across chain length: the **bioregulators** (Khavinson “cytomedines”) – very short di-/tripeptides claimed to regulate tissue-specific genes. Examples here include **Epitalon** and **Pinealon** (with Cortagen and Vesugen). Defined by mechanism rather than structure, the category rests on one Russian group without independent Western replication – as the GRADE C/D ratings reflect.

Routes of administration

Route selection is determined by molecular weight, gut stability, target tissue, and the half-life the formulation needs to achieve.

ROUTE	BIOAVAILABILITY	CLINICAL REALITY
Subcutaneous (SC)	~70–90%	The default for most peptide drugs. Self-administered with a 29–31 gauge insulin syringe; rotate injection sites to avoid lipohypertrophy.
Intramuscular (IM)	~75–95%	Faster onset than SC, more painful. Used for depot formulations (leuprolide, lanreotide), emergency glucagon, and a few clinic-administered peptides.
Intravenous (IV)	100%	Clinical setting only — insulin for DKA, glucagon for hypoglycemia, octreotide infusions for variceal bleeding.
Oral	< 1% native	Most peptides are destroyed in the gut. Successful oral peptides either use absorption enhancers (SNAC for oral semaglutide) or are not meant to be absorbed (linaclotide acts in the gut lumen).
Intranasal	~5–30%	Useful for calcitonin, desmopressin, oxytocin, Selank, Semax. Direct CNS access via the olfactory nerve makes intranasal attractive for neuropeptides.
Sublingual / buccal	Variable	Bypasses first-pass metabolism. Limited by peptide molecular weight and lipophilicity; mostly research-grade preparations.
Topical	Local only	Skin barrier limits systemic absorption. GHK-Cu cosmetics, BPC-157 ointments. Local effects only — claims of systemic activity from topical application are usually unsupported.

Reconstitution & storage

RECONSTITUTION

Most injectable peptides ship as lyophilized (freeze-dried) powder that must be reconstituted before use. The mechanics matter — incorrect technique can denature 20–40% of the peptide before the first dose is drawn.

Reconstitution calculator: visit www.hormonal.ai/calculator to compute reconstitution volumes, syringe units (U-100), and dose conversions for any peptide vial — useful for verifying that the volume + concentration combination gives a clean dose draw on the syringe scale you have.

- **Solvent.** Bacteriostatic water for injection (BWFI, 0.9% benzyl alcohol) for multi-dose vials; sterile water for single-use only.
- **Technique.** Inject solvent slowly down the inside wall of the vial. Do not aim the stream at the powder. Swirl gently — do not shake. Shaking creates foam and can denature the peptide.
- **Concentration.** Pick a reconstitution volume that gives convenient dosing. A 5 mg vial reconstituted in 2.5 mL BWFI yields 2 mg/mL; a 250 µg dose is then 0.125 mL, or 12.5 units on a U-100 insulin syringe.
- **Pre-dose inspection.** The reconstituted solution should be clear and colourless. Any cloudiness, particulate matter, or yellowing indicates the peptide has degraded — discard and start a new vial.

STORAGE RULES

- **Lyophilized vial:** refrigerate at 2–8 °C. Frozen (–20 °C) for long-term storage. Many peptides tolerate room temperature for weeks if dry and well-sealed.
- **Reconstituted vial (BWFI):** refrigerate at 2–8 °C; use within ~28 days. The benzyl alcohol provides bacteriostatic protection but does not prevent peptide degradation.
- **Reconstituted vial (sterile water):** refrigerate and use within 24–48 hours. No preservative against bacterial growth.
- **Light and heat:** store in original container, away from direct light. Avoid leaving in a hot car, a window, or near radiators. Cold-chain shipping with ice packs preserves the molecule; ambient overnight shipping in summer can degrade it noticeably.
- **Travel:** for travel, transfer vials to an insulated bag with refrigerated ice packs. A 24-hour ambient exposure is generally tolerated; longer requires re-evaluation.

Stability & degradation

Peptides are intrinsically unstable molecules. Knowing the degradation pathways tells you how to store, when to discard, and what to look for in the vial.

THE FIVE DEGRADATION PATHWAYS

- **Oxidation** of methionine, cysteine, and tryptophan residues. Mitigated by antioxidants and oxygen-free packaging.
- **Hydrolysis** of peptide bonds in solution — accelerated by extreme pH, heat, and time. Lyophilized form is dramatically more stable.
- **Deamidation** of asparagine and glutamine to acidic forms — a major cause of slow potency loss in stored solutions, occurring even refrigerated.
- **Aggregation** — peptides dimerize, oligomerize, or precipitate at high concentration. Surfactants and gentle handling help.
- **Adsorption** — peptides stick to glass and plastic surfaces, silently depleting low-concentration solutions over weeks.

VISUAL SIGNS OF DEGRADED PRODUCT

Cloudy or particulate solution after reconstitution · yellowing of the powder or solution · loss of clinical effect at a previously effective dose · precipitate that fails to redissolve on gentle warming · off-smell from the vial (rare but a hard signal).

PREFILLED PEN STABILITY

Prefilled pens are factory-formulated with preservatives and tested under cGMP. The manufacturer-validated in-use stability of the major GLP-1 pens, per FDA-approved labelling:

- **Ozempic and Wegovy (semaglutide):** 56 days at controlled room temperature (15–30 °C / 59–86 °F) after first use; refrigeration permitted but does not extend the window.
- **Mounjaro KwikPen (tirzepatide, multi-dose):** 30 days at room temperature once in use. Single-dose Mounjaro and Zepbound pens: 21-day room-temperature limit before use, then single injection and disposal.
- **Compounded semaglutide and tirzepatide:** pharmacy-assigned beyond-use dates (BUDs) governed by USP <797>, typically shorter than the labelled in-use windows for branded pens.
- **Gray-market pens and vials:** sold without manufacturer stability validation, cGMP oversight, or independent testing. Stability is undocumented and unverifiable; the most conservative of the above limits should be presumed at best.

Manufacturing

SOLID-PHASE PEPTIDE SYNTHESIS (SPPS)

Invented by Bruce Merrifield in 1963 (Nobel Prize 1984). The peptide is built one amino acid at a time on a polymer resin. Each cycle deprotects the chain, couples the next Fmoc-protected amino acid, and washes — typically 30–60 minutes per residue. After assembly, the peptide is cleaved from resin, side-chains deprotected, and the crude product purified by reverse-phase HPLC.



Yields fall geometrically: 98% per coupling step $\times$ 50 cycles $\approx$ 36% overall yield. The practical ceiling for routine pharmaceutical manufacture is $\sim$ 50 residues. Beyond that, recombinant approaches become more economical.

HALF-LIFE EXTENSION STRATEGIES

STRATEGY	EXAMPLE	$t_{1/2}$
Fatty-acid acylation	Semaglutide (C18 diacid)	$\sim$7 days
Fc fusion	Dulaglutide (IgG4 Fc)	$\sim$5 days
Depot formulation	Lanreotide autoquel	$\sim$28 days

RECOMBINANT DNA TECHNOLOGY

For longer peptides and proteins. The peptide-coding DNA is inserted into a host organism — *E. coli* (insulin), *Pichia pastoris* (some calcitonin), or mammalian CHO cells (Fc-fusion drugs). Used for insulin, growth hormone, and most GLP-1 analogues; the recombinant backbone is then chemically conjugated with fatty acids or other half-life extenders.

Regulatory grades

The same molecule can exist in five radically different regulatory states. Understanding the hierarchy is essential before prescribing or recommending any peptide.



FDA

FDA-approved (or equivalent). Full Phase 3 trials completed; approved label, indications, dosing, contraindications. Manufactured to cGMP; post-market surveillance active; insurance reimbursable. Examples: insulin, semaglutide, tesamorelin, leuprolide, octreotide. About half the peptides in this guide.

503B

503B outsourcing facility. Compounded under cGMP for office-use stock. Registered with the FDA and inspected. The compound itself is not FDA-approved — only the manufacturing standard is. Used for shortages, off-label needs, or patient-specific requirements. Quality bar is high; documentation is comprehensive.

503A

503A compounding pharmacy. State-licensed pharmacy compounding for a specific patient prescription; lower bar than 503B. Must use bulk substances on the FDA compounding list, a USP/NF monograph, or an approved drug product. In September 2023 the **FDA 503A Category 2 list** identified certain bulk substances — including several peptides (BPC-157, TB-500, melanotan II, and others) — as presenting significant safety risks for compounding, pending PCAC review.

RUO

Research Use Only. Sold by chemical-supply companies (Bachem, Sigma, Anaspec, others) labeled "for research only — not for human consumption." Purity varies from pharmaceutical-grade (Bachem) to unverified (smaller suppliers). Outside pharmaceutical regulation. The disclaimer provides no legal protection — the FDA regulates based on inferred intended use.

GRAY

Gray market / underground. "Peptide research" websites with deliberate ambiguity. Quality ranges from acceptable to dangerous. May contain wrong peptides, sub-potent material, bacterial endotoxin, residual TFA, or heavy metals. No regulatory oversight. Independent testing has found peptide content from 60–110% of stated amount.

Reading a Certificate of Analysis

A COA is the supplier's quality attestation. A serious COA tells you the molecule is what it claims, how pure it is, what counter-ion it carries, and whether it is safe to inject.

IDENTITY

Mass spectrometry confirms the molecular weight matches the expected peptide. The theoretical monoisotopic mass should be reported alongside the observed mass. Deviations > 1 Da indicate sequence problems.

WATER CONTENT

Karl Fischer titration. Lyophilized peptides typically retain 3–10% residual moisture. Higher water content reduces stability.

ENDOTOXIN (LAL)

For injectable use, endotoxin must be < 0.5 EU/mg by Limulus Amebocyte Lysate assay. RUO peptides frequently have no endotoxin testing — a hard contraindication to injection.

HEAVY METALS

Lead, cadmium, mercury, arsenic by ICP-MS. Within ICH Q3D limits.

PURITY

HPLC chromatogram with retention time and peak integration. Pharmaceutical grade is ≥ 98%; RUO is typically 95–99%; below 95% is significant impurity load.

COUNTER-ION (TFA / ACETATE)

Synthetic peptides ship as TFA or acetate salts. A 5 mg "peptide" vial may contain 4.2 mg net peptide after subtracting the counter-ion. Pharmaceutical preparations always report this.

MICROBIAL CONTAMINATION

Total aerobic count, yeast and mold count, absence of pathogens. Critical for any injectable preparation.

RESIDUAL SOLVENTS

DMF, DMSO, methanol from SPPS. Within ICH Q3C limits.

Quality red flags

WHAT YOU SHOULD NOT SEE ON A SERIOUS COA

- A bare claim of "purity > 99%" with no chromatogram or supporting data.
- No mass spectrum, or a spectrum without a labelled expected mass.
- No batch or lot number; no manufacture date.
- A COA dated more than 12 months before purchase — peptides degrade in storage.
- Generic identical-looking COAs across different products from the same vendor.
- No endotoxin or microbial data on a product marketed for injection.
- A COA referring to a "typical" batch rather than the specific batch shipped.

Practical heuristic. Legitimate peptide suppliers attach a lot-specific COA to every vial sold and respond to documentation requests within hours. If you must request a COA and the response is delayed, generic, or never arrives, the product is probably not what it claims to be — and certainly not safe to inject.

FOR 503A/503B COMPOUNDED PEPTIDES

Compounding pharmacies should provide a COA documenting identity, purity, sterility, and endotoxin. Ask which testing laboratory ran the analysis, the date of testing, the USP-797 / 800 compliance status of the facility, and the lot number that links the COA to the vial in hand. A pharmacy that cannot produce this documentation on request is not operating to the standards their license requires.

FOR RESEARCH-USE-ONLY PEPTIDES

If the product is being used outside its labelled purpose, the standard of evidence shifts to the user. At minimum: confirm identity by independent mass spectrometry, confirm purity by independent HPLC, run an endotoxin test before any injectable use, and document the source, lot, and date for every dose. None of this turns RUO into pharmaceutical grade, but it surfaces problems before they reach a patient.

THERAPEUTIC CLASS

Metabolic Peptides

“Incretin biology reshaped obesity medicine more rapidly than any endocrine therapeutic class in modern history — drugs discovered by studying Gila monster venom now govern the metabolism of hundreds of millions.”

10 ENTRIES

Insulin

Humulin · Novolin · Lantus · Levemir · Tresiba · Humalog · NovoLog · Apidra

MECHANISM

Binds the insulin receptor (a tyrosine kinase), triggering GLUT4 translocation to the cell membrane and driving glucose uptake; suppresses hepatic gluconeogenesis.

INDICATIONS

- Type 1 diabetes
- Type 2 diabetes (when oral therapy insufficient)
- Diabetic ketoacidosis
- Hyperkalemia (with dextrose)

DOSING

Type 1 DM total daily	0.5–1.0 U/kg/day, basal-bolus split
Long-acting basal	1–2×/day
Rapid-acting prandial	Per meal, weight- and carb-counted

EVIDENCE Foundational therapy with 100+ years of use. Modern analogues have extensive RCT evidence for glycemic control and outcome benefit.

SAFETY Hypoglycemia is the principal acute risk · Weight gain · Lipohypertrophy at injection sites

ADVERSE Hypoglycemia (most common, dose-dependent) · weight gain · injection-site lipohypertrophy or lipoatrophy · local hypersensitivity at injection sites · rare systemic allergy. Severe hypoglycemia is the principal serious AE.

CLINICAL CAVEAT Modern analogues (glargine, degludec, lispro, aspart) have replaced NPH and regular insulin for most uses — choice is by pharmacokinetic profile, not generic class.

KEY REFERENCES

- Banting FG, Best CH. J Lab Clin Med 1922;7:251
- DCCT Research Group. NEJM 1993;329:977 (PMID 8366922)
- UKPDS 33. Lancet 1998;352:837 (PMID 9742976)

FDA-approved · all major regulators

GRADE A

GLP-1 RECEPTOR AGONIST

Semaglutide

Ozempic · Wegovy · Rybelsus

MECHANISM

Long-acting GLP-1 receptor agonist; albumin-bound fatty-acid chain extends half-life to ~7 days. Slows gastric emptying, augments glucose-dependent insulin release, suppresses glucagon, and reduces appetite via central GLP-1 receptors.

INDICATIONS

- Type 2 diabetes
- Chronic weight management (BMI ≥ 30 , or ≥ 27 with comorbidities)
- Reduce MACE in T2DM with established CVD
- Reduce MACE in adults with obesity and established CVD (Wegovy, SELECT)

DOSING

SC weekly (T2DM/obesity)	0.25 mg start; titrate to 2.0 mg (T2DM) or 2.4 mg (obesity)
Oral (Rybelsus)	3 mg → 7 mg → 14 mg daily, 30-day titration; fasted, no food/meds 30 min after

EVIDENCE STEP 1 (n=1961): 14.9% mean weight loss vs 2.4% placebo at 68 wks (Wilding 2021). SUSTAIN program established T2DM glycemic efficacy. SELECT trial (2023): ~20% MACE reduction in established CVD without diabetes.

SAFETY Nausea, vomiting (titration-dependent) · Pancreatitis risk (rare) · Contraindicated in personal/family history of MTC or MEN2 · Diabetic retinopathy progression signal

ADVERSE Most common: nausea (44%), diarrhea (32%), vomiting (24%), constipation (24%), abdominal pain (20%). Less common: AKI (volume-depletion).

CLINICAL CAVEAT Titrate slowly (4-wk increments) to limit nausea-driven discontinuation. Delayed gastric emptying alters absorption of narrow-window oral drugs. See appendix: neuropsychiatric and addiction signals in GLP-1 therapy.

KEY REFERENCES

- Wilding JPH et al. STEP 1. NEJM 2021;384:989 (PMID 33567185)
- Lincoff AM et al. SELECT. NEJM 2023;389:2221 (PMID 37952131)
- Marso SP et al. SUSTAIN-6. NEJM 2016;375:1834 (PMID 27633186)

FDA-approved · EMA · most major regulators

GRADE A

GIP / GLP-1 DUAL AGONIST

Tirzepatide

Mounjaro · Zepbound

MECHANISM

Activates both GIP and GLP-1 receptors. The dual agonism produces greater appetite suppression and improved insulin sensitivity than GLP-1 monotherapy.

INDICATIONS

- Type 2 diabetes
- Chronic weight management (BMI ≥ 30 or ≥ 27 with comorbidity)
- Obstructive sleep apnea (Zepbound, 2024)

DOSING

SC weekly	2.5 mg → titrate by 2.5 mg every 4 weeks
Max dose	15 mg weekly
OSA dose	10–15 mg weekly
Microdosing (off-label)	0.5–2.5 mg SC weekly (compounded) · <i>not FDA-validated</i>

EVIDENCE SURMOUNT-1 (Jastreboff 2022): 22.5% mean weight loss at 15 mg over 72 wks. SURPASS-2: HbA1c reduction of 2.46% (tirzepatide 15 mg) vs 1.86% (semaglutide 1 mg). Superior weight-loss and glycemic efficacy across the program.

SAFETY GI: nausea, vomiting, diarrhea · Pancreatitis risk · MTC/MEN2 contraindication · Gallbladder disease signal

ADVERSE Most common: nausea (29%), diarrhea (23%), decreased appetite (11%), vomiting (10%), constipation, dyspepsia. Less common: modest heart-rate rise, hypoglycemia with insulin or SU.

CLINICAL CAVEAT SURPASS-2 head-to-head: greater weight loss and A1C reduction vs semaglutide 1 mg; the driver (dual GIP/GLP-1 vs dose/PK) is unresolved. Compounded microdosing (0.5–2.5 mg/wk) is off-label, not FDA-validated.

KEY REFERENCES

- Jastreboff AM et al. SURMOUNT-1. NEJM 2022;387:205 (PMID 35658024)
- Frias JP et al. SURPASS-2. NEJM 2021;385:503 (PMID 34170647)
- Malhotra A et al. SURMOUNT-OSA. NEJM 2024;391:1193 (PMID 38912654)

FDA-approved · EMA · most major regulators

GRADE A

SMALL-MOLECULE (NON-PEPTIDE) ORAL GLP-1 RA

Orforglipron

Foundayo (Eli Lilly · LY3502970)

MECHANISM

Once-daily oral non-peptide GLP-1R agonist. Full-agonist signaling; no food/water restrictions (unlike Rybelsus). Same downstream pharmacology as peptide GLP-1 RAs — augmented insulin release, glucagon suppression, slowed gastric emptying, central appetite reduction.

INDICATIONS

- Chronic weight management (obesity / overweight + comorbidity)
- Investigational: type 2 diabetes (ACHIEVE program), OSA, knee OA, hypertension

DOSING

Oral daily (obesity, approved)	6 / 12 / 36 mg daily; no food or water restriction
Oral daily (T2D · trial dosing)	3 / 12 / 36 mg in Phase 3 — T2D approval pending · <i>not FDA-validated</i>

EVIDENCE Multiple positive Phase 3 readouts. ATTAIN-1 (NEJM 2025): 12.4% weight loss at 36 mg / 72 wks. ATTAIN-2 (T2D+obesity): 10.5% weight loss, A1C -1.8%. ACHIEVE-1: A1C -1.3% to -1.6%. ACHIEVE-3 (Lancet 2026): beat oral semaglutide head-to-head on A1C by 0.44–0.48 ppts at 52 wks.

SAFETY GI: nausea, diarrhea, vomiting consistent with injectable GLP-1 class · No new safety signals vs injectable GLP-1 RAs in Phase 3 · Discontinuation rate comparable to oral semaglutide

ADVERSE Most common: nausea, diarrhea, vomiting, constipation, decreased appetite — consistent with injectable GLP-1 class.

CLINICAL CAVEAT First FDA-approved oral non-peptide GLP-1 RA (Foundayo). Major practical advantage over Rybelsus: no food/water restrictions. Beat oral semaglutide head-to-head in ACHIEVE-3. T2D indication pending. Use with other GLP-1 RAs not recommended; same boxed thyroid C-cell tumor warning as injectable class.

KEY REFERENCES

- Wadden TA et al. ATTAIN-1 (Phase 3 obesity). NEJM 2025 (NCT05869903)
- ATTAIN-2 (Phase 3 T2D + obesity). Topline 2025
- ACHIEVE-1 (Phase 3 T2D). Topline 2025

FDA-approved (Foundayo, Apr 2026) · obesity · T2D pending

GRADE B

GIP / GLP-1 / GLUCAGON TRIPLE AGONIST

Retatrutide

Investigational (Eli Lilly)

MECHANISM

Co-activates GIP, GLP-1, and glucagon receptors. Glucagon agonism raises hepatic energy expenditure, shifting the weight-loss balance relative to GIP/GLP-1 dual agonists.

INDICATIONS

- Obesity (investigational)
- Type 2 diabetes (investigational)
- MASH (investigational)

DOSING · reported-use only, not recommended dosing; no approved indication

Phase 3 obesity (trial dosing) 4, 9, or 12 mg SC weekly; 12 mg achieved 28.3% weight loss at 80 weeks (TRIUMPH-1); no approved dose

EVIDENCE Phase 2 (Jastreboff 2023, n=338): 24.2% weight loss at 48 wks (12 mg). MASLD substudy (Sanyal 2024): 86% liver-fat reduction. TRIUMPH-4 Phase 3 (Dec 2025): 28.7% at 68 wks (obesity + knee OA). TRIUMPH-1 (May 2026, n=2339): 28.3% at 80 wks, 30.3% at 104 wks — pivotal obesity trial.

SAFETY Same incretin GI profile (nausea, vomiting) · Glucagon component raises modest concern for hepatic glucose output; no clinically apparent issue in Phase 2 · Heart rate increase noted

ADVERSE Phase 2/3: nausea (most common, dose-dependent), vomiting, diarrhea, constipation, decreased appetite. Modest heart-rate rise (~5 bpm).

CLINICAL CAVEAT The glucagon component adds hepatic energy-expenditure effects to incretin-mediated appetite suppression; whether this yields a clinically meaningful edge over tirzepatide awaits head-to-head data. “GLP-3” is an informal online nickname — retatrutide is a triple GLP-1/GIP/glucagon agonist. Graded B (investigational): strong Phase 3 data but not yet approved or standard of care.

KEY REFERENCES

- Jastreboff AM et al. Phase 2 obesity. NEJM 2023;389:514 (PMID 37366315)
- Rosenstock J et al. Phase 2 T2DM. Lancet 2023;402:529 (PMID 37385275)
- Sanyal AJ et al. Phase 2a MASLD substudy. Nat Med 2024;30:2037 (DOI 10.1038/s41591-024-03018-2)

Investigational · TRIUMPH Phase 3 readouts positive · FDA filing expected late-2026/2027

GRADE A

GLP-1 RECEPTOR AGONIST

Liraglutide

Victoza · Saxenda

MECHANISM

Daily-injection GLP-1 receptor agonist with ~13 hour half-life via albumin binding. Same mechanism as semaglutide; shorter duration.

INDICATIONS

- Type 2 diabetes
- Chronic weight management in adults
- Pediatric obesity (ages 12+)

DOSING

Daily SC (T2DM) 0.6 mg → 1.2 → 1.8 mg

Daily SC (obesity) 0.6 mg → titrate weekly to 3.0 mg

EVIDENCE LEADER trial showed MACE reduction in T2DM with CV disease. SCALE trial showed ~8% mean weight loss at 56 wks.

SAFETY GI side effects (typically dose-dependent) · Pancreatitis risk · MTC/MEN2 contraindication

ADVERSE Most common: nausea (39%), diarrhea (21%), constipation (19%), vomiting (16%), headache, fatigue. Less common: gallbladder disease, acute pancreatitis, hypoglycemia (with insulin/SU), injection-site reactions.

CLINICAL CAVEAT Largely displaced by once-weekly agents in adults; remains the only GLP-1 approved for pediatric obesity (ages 12+). See appendix: neuropsychiatric and addiction signals in GLP-1 therapy.

KEY REFERENCES

- Pi-Sunyer X et al. SCALE Obesity. NEJM 2015;373:11 (PMID 26132939)
- Marso SP et al. LEADER. NEJM 2016;375:311 (PMID 27295427)
- Davies MJ et al. SCALE Diabetes. JAMA 2015;314:687 (PMID 26284720)

FDA-approved (Victoza 2010, Saxenda 2014) · EMA

GRADE A

GLP-1 RECEPTOR AGONIST

Exenatide

Byetta · Bydureon

MECHANISM

GLP-1 receptor agonist; resistant to DPP-4 cleavage due to alanine-to-glycine substitution at position 2.

INDICATIONS

— Type 2 diabetes (adjunct)

DOSING

SC 2×/day (Byetta) 5 µg → 10 µg 2×/day

SC weekly (Bydureon) 2 mg weekly

EVIDENCE EXSCEL trial showed neutral cardiovascular outcomes in T2DM. Efficacy generally lower than semaglutide/tirzepatide.

SAFETY GI side effects · Injection-site nodules with Bydureon (depot) · Pancreatitis risk · MTC contraindication

ADVERSE Most common: nausea (44%), vomiting (13%), diarrhea (13%), feeling jittery, dizziness, headache, dyspepsia. Bydureon depot: injection-site nodules (long-lasting), pruritus. Less common: acute pancreatitis, AKI.

CLINICAL CAVEAT Twice-daily Byetta is now rarely used; weekly Bydureon depot creates persistent injection-site nodules in roughly 17% of patients. EXSCEL was CV-neutral, unlike semaglutide and liraglutide. See appendix: neuropsychiatric and addiction signals in GLP-1 therapy.

KEY REFERENCES

- Buse JB et al. AMIGO-1. Diabetes Care 2004;27:2628 (PMID 15504997)
- Holman RR et al. EXSCEL. NEJM 2017;377:1228 (PMID 28910237)

FDA-approved (Byetta 2005, Bydureon 2012) · EMA

GRADE A

GLP-1 RECEPTOR AGONIST

Dulaglutide

Trulicity

MECHANISM

GLP-1 receptor agonist; IgG4 Fc fragment extends half-life to ~5 days via FcRn recycling.

INDICATIONS

- Type 2 diabetes
- MACE reduction in T2DM ± established CVD

DOSING

SC weekly 0.75 mg → 1.5 mg → up to 4.5 mg

EVIDENCE REWIND trial: significant MACE reduction in 9,901 patients with T2DM at lower CV risk than prior trials.

SAFETY GI side effects · Pancreatitis risk · MTC/MEN2 contraindication

ADVERSE Most common: nausea (12–21%), diarrhea (12–13%), vomiting (6–13%), abdominal pain, decreased appetite, dyspepsia, fatigue. Less common: pancreatitis, hypersensitivity reactions, hypoglycemia (with insulin/SU).

CLINICAL CAVEAT Single-use pen makes administration easier than semaglutide for elderly or dexterity-limited patients. REWIND showed CV benefit in lower-risk T2DM than other GLP-1 outcome trials. See appendix: neuropsychiatric and addiction signals in GLP-1 therapy.

KEY REFERENCES

- Gerstein HC et al. REWIND. Lancet 2019;394:121 (PMID 31189511)
- Probst L et al. JCI Insight 2023;8:e170419 (PMID 37971184)

FDA-approved (Trulicity, 2014) · EMA

GRADE B

AMYLIN ANALOGUE

Pramlintide

Symlin

MECHANISM

Activates the amylin/CTR-RAMP receptor complex. Slows gastric emptying, suppresses glucagon, and reduces postprandial glucose excursions.

INDICATIONS

- Type 1 diabetes (adjunct to insulin)
- Type 2 diabetes (with insulin)

DOSING

Premeal SC (T1DM) 15 µg → 60 µg per meal

Premeal SC (T2DM) 60 µg → 120 µg per meal

EVIDENCE Multiple RCTs show modest A1c reduction (~0.4%) and modest weight loss as adjunct to insulin.

SAFETY Severe hypoglycemia risk when combined with insulin (mandate insulin dose reduction) · Nausea common · Boxed warning for severe insulin-induced hypoglycemia

ADVERSE Severe insulin-induced hypoglycemia within 3 hours of co-administration (boxed warning). Nausea (28–48%), anorexia (10–17%), vomiting, headache, fatigue, abdominal pain. Mandates insulin dose reduction.

CLINICAL CAVEAT Cannot be mixed with insulin in the same syringe — requires a separate injection. Insulin dose must be reduced by 50% at initiation to prevent severe hypoglycemia.

KEY REFERENCES

- Whitehouse F et al. Diabetes Care 2002;25:724 (PMID 11919132)
- Hollander PA et al. Diabetes Care 2003;26:784 (PMID 12610038)

FDA-approved · Symlin (AstraZeneca) · 2005

GRADE A

PANCREATIC COUNTER-REGULATORY HORMONE

Glucagon

GlucaGen · Baqsimi · Gvoke

MECHANISM

Activates the hepatic glucagon receptor (Gs-coupled GPCR), driving glycogenolysis and gluconeogenesis to raise blood glucose.

INDICATIONS

- Severe hypoglycemia (emergency)
- Diagnostic agent for GI motility imaging

DOSING

IM/SC emergency	1 mg (adult), 0.5 mg (pediatric <25 kg) — Gvoke autoinjector, GlucaGen vial
Intranasal emergency	3 mg single dose, one nostril — Baqsimi
Diagnostic	0.25–2 mg IV

EVIDENCE Standard-of-care emergency therapy; modern formulations show equivalent efficacy with much improved ease of use.

SAFETY Nausea, vomiting after administration · Rebound hyperglycemia · Contraindicated in pheochromocytoma, insulinoma

ADVERSE Nausea and vomiting after administration (very common), hyperglycemia (expected effect), transient hypokalemia, headache. Rare: hypersensitivity, transient hypertension in patients with pheochromocytoma.

CLINICAL CAVEAT Newer formulations (Baqsimi nasal, Gvoke ready-to-use pen) eliminated reconstitution at the bedside, simplifying out-of-hospital hypoglycemia rescue.

KEY REFERENCES

- Pieber TR et al. Baqsimi PK/PD. Diabetes Care 2015
- Christiansen MP et al. Gvoke. Diabetes Care 2019 (PMID 30765431)

FDA-approved · all major regulators

THERAPEUTIC CLASS

Growth Hormone Axis

“The somatotrophic axis governs not only growth and body composition but serves as a molecular clock of aging — explaining why its pharmacology attracts both endocrinologists and longevity researchers.”

10 ENTRIES

GRADE A

RECOMBINANT 191-AMINO-ACID GROWTH HORMONE

Somatropin (hGH)

Genotropin · Norditropin · Humatrope · Saizen · Omnitrope (+ weekly forms)

MECHANISM

Binds the dimeric GH receptor (JAK2/STAT5), driving hepatic and peripheral IGF-1 production plus direct lipolytic, anabolic, and insulin-antagonizing effects. Somatropin is identical to the 191-residue pituitary hormone.

INDICATIONS

- Approved: pediatric & adult GHD, Turner, Prader-Willi, SGA, ISS, HIV wasting, short bowel
- Off-label: anti-aging, body composition, bodybuilding (unapproved; WADA-prohibited)

DOSING

SC daily (adult GHD)	0.2–0.4 mg/day, titrate to IGF-1; long-acting weekly forms also approved
Pediatric (approved)	0.16–0.7 mg/kg/week divided daily, indication-dependent
Off-label/bodybuilding (gray-market)	2–6 IU/day SC ($\approx$ 0.7–2 mg; 1 mg $\approx$ 3 IU); supraphysiologic · <i>not FDA-validated</i>

EVIDENCE Decades of RCTs establish efficacy for approved GH-deficiency and growth indications. The small uncontrolled TRIIM trial (Fahy 2019, n=9) paired rhGH with DHEA and metformin and reported thymic regeneration and a mean epigenetic age $\sim$ 2.5 years below untreated at 1 year — hypothesis-generating, not proof.

SAFETY Edema, arthralgia, myalgia, carpal tunnel · Insulin resistance, hyperglycemia; intracranial hypertension · Contraindicated in active malignancy and acute critical illness

ADVERSE Most common: fluid retention, arthralgia, myalgia, carpal-tunnel symptoms, glucose intolerance. Supraphysiologic abuse causes acromegaly-like changes, diabetes, and a theoretical IGF-1-driven cancer concern.

CLINICAL CAVEAT Approved replacement for documented GH deficiency, dosed to IGF-1 — not a validated anti-aging or performance drug; supraphysiologic off-label/bodybuilding use is unapproved and WADA-prohibited. The TRIIM epigenetic-age result (Fahy 2019) came from a tiny uncontrolled GH/DHEA/metformin cohort, so the GH-specific effect is unproven.

KEY REFERENCES

- Molitch ME et al. Adult GHD guideline. JCEM 2011;96:1587 (PMID 21602453)
- Fahy GM et al. TRIIM. Aging Cell 2019;18:e13028 (PMID 31496122)

FDA-approved (somatropin) · WADA-prohibited · off-label/bodybuilding gray-market

GRADE A

GHRH ANALOGUE

Tesamorelin

Egrifta SV · Egrifta WR

MECHANISM

GHRH receptor agonist on pituitary somatotrophs; stimulates pulsatile endogenous growth hormone release. Lipolysis in visceral fat appears to be a downstream effect of restored GH/IGF-1 axis.

INDICATIONS

— HIV-associated lipodystrophy (visceral adiposity reduction)

DOSING

SC daily (approved) 1.4 mg SC daily (Egrifta SV, 2 mg/vial) or 1.28 mg SC daily (Egrifta WR, 11.6 mg/vial)

Off-label 1 mg SC daily for visceral fat reduction and GH/IGF-1 optimization · *not FDA-validated*

EVIDENCE Pivotal Phase 3 trials (Falutz 2007) showed ~15% reduction in visceral adipose tissue at 26 wks vs placebo, with concomitant IGF-1 elevation into the upper-normal range. Effect sustained over 52-week extension.

SAFETY Injection site reactions common · Fluid retention · Glucose intolerance · Contraindicated in active malignancy

ADVERSE Most common: injection-site reactions (erythema, pruritus, pain, irritation, swelling — 25%). Arthralgia, peripheral edema, myalgia, paresthesia. Glucose intolerance and fluid retention with chronic use. Hypersensitivity reactions reported.

CLINICAL CAVEAT Tesamorelin is the only FDA-approved GHRH analogue; sermorelin, CJC-1295, and ipamorelin all lack equivalent regulatory backing despite chemically similar GH-axis profiles. Off-label gray-market practice frequently combines tesamorelin with ipamorelin (CJC/lpa-style stacking) to layer GHRH-driven GH release with GHSR-driven pulses — no FDA endorsement of this combination and not on-label for any indication.

KEY REFERENCES

- Falutz J et al. NEJM 2007;357:2359 (PMID 18057338)
- Stanley TL et al. JAMA 2014;312:380 (PMID 25058085)

FDA-approved (Egrifta, HIV lipodystrophy, 2010) · EU application withdrawn 2012

GRADE B

GHRH(1-29) ANALOGUE

Sermorelin

Geref (discontinued) · compounded

MECHANISM

GHRH receptor agonist; identical mechanism to tesamorelin but with shorter half-life (~10 min). Promotes pulsatile endogenous GH release that remains subject to somatostatin negative feedback — making overdose difficult, unlike exogenous HGH.

INDICATIONS

- Historical: pediatric GH deficiency
- Off-label adult use: marketed for age-related GH/IGF-1 decline (no on-label adult indication)

DOSING · *reported-use only, not recommended dosing; no approved indication*

Pediatric (historical)	30 µg/kg SC at bedtime
Adult off-label	100–300 µg SC at bedtime, 5–7 nights/week
Compounded blends	Often paired with ipamorelin or GHRP-6

EVIDENCE Pediatric efficacy well-established in the pre-approval trials. Adult data are limited to small RCTs (n=10–19, 16 wks) showing lean-mass gains (+1.26 kg in men), improved insulin sensitivity, and raised IGF-1.

SAFETY Injection-site reactions · Headache, flushing · Risk of glucose intolerance with long-term use

ADVERSE Injection-site reactions (erythema, pain — most common), headache (rare), flushing, dizziness, urticaria, hyperactivity, sleep disturbance. Generally well tolerated in pediatric and adult use.

CLINICAL CAVEAT Commonly paired with ipamorelin in compounded blends (combined GHRH + GHSR stimulation). Key contrast: tesamorelin remains FDA-approved with rigorous manufacturing QC, whereas sermorelin is now compounding-only without FDA quality oversight. Banned by most athletic organizations; adult anti-aging use remains poorly studied.

KEY REFERENCES

- Vittone J et al. Metabolism 1997;46:89 (PMID 9141536)
- Walker RF et al. Pediatrics 1991
- Aleppo G et al. (review) Endocr Pract 1997

Compounded only · pediatric brand (Geref) discontinued 2008 · never approved for adult use

GRADE C

GHRH ANALOGUE (MODIFIED)

CJC-1295

Research only · compounded

MECHANISM

GHRH(1-29) analogue with four amino-acid substitutions that resist protease (DPP-4) cleavage. The DAC variant adds a maleimidopropionic acid-lysine linker that covalently binds serum albumin in vivo, extending half-life from ~30 min (native GHRH) to 5–8 days and enabling weekly dosing. The sustained signal gives continuous rather than pulsatile GH stimulation — arguably less physiologic than short-acting GHRH analogues.

INDICATIONS

- None approved
- Research and gray-market off-label use: marketed for age-related GH/IGF-1 decline

DOSING · reported-use only, not recommended dosing; no approved indication

Phase 2 trial range (with DAC)	30–60 µg/kg SC 1–2×/week
Compounded with DAC	1–2 mg SC weekly (typical anti-aging clinic use)
Without DAC ("Mod GRF")	100 µg SC 1–2×/day

EVIDENCE Phase 2 (with DAC, 30–60 µg/kg SC): mean GH rose 2–10× and IGF-1 1.5–3× for 6–14 days after a single injection. Human development was discontinued after a participant death during a Phase 2 trial — attributed to cardiovascular causes and not definitively linked to the drug. No positive efficacy RCT; no Phase 3.

SAFETY Sustained high IGF-1 raises theoretical concern for malignancy promotion · Injection-site reactions · Glucose intolerance possible · Not approved anywhere

ADVERSE Limited human data. Reported: injection-site reactions, flushing, transient headache, mild water retention, numbness/tingling.

CLINICAL CAVEAT Evidence base: Phase 1/2 safety pharmacology and animal models plus extensive off-label use — no positive human efficacy RCTs. Two forms: Mod GRF (no DAC, daily, pulsatile) vs CJC-1295 with DAC (weekly, sustained IGF-1); the DAC version raises theoretical proliferation concerns absent with the non-DAC form. No Phase 3 or current pharma development.

KEY REFERENCES

- Teichman SL et al. JCEM 2006;91:799 (PMID 16352683)
- Ionescu M, Frohman LA. JCEM 2006;91:4792 (PMID 16985089)

Research only · gray-market

GRADE C

GH SECRETAGOGUE (GHSR AGONIST)

Ipamorelin

Research only · compounded

MECHANISM

Synthetic pentapeptide ghrelin mimetic; selective GHS-R1a agonist that releases GH and is reported to preserve the natural pulsatile pattern. Unlike GHRP-6, GHRP-2, and hexarelin, it raises cortisol, prolactin, and ACTH minimally.

INDICATIONS

- None approved
- Off-label/gray-market: body composition, recovery, sleep, collagen synthesis (anti-aging & sports clinics)

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label SC

100–300 µg SC 1–2×/day (start 200 µg); empty stomach (fast 2 h before, 30 min after); pre-bed amplifies the nocturnal GH pulse

Cycling/stacking

8–12 weeks on, 4 weeks off; often combined with CJC-1295 (no DAC); do not stack other GH secretagogues

EVIDENCE Raun 1998 (swine): dose-dependent GH release with high selectivity. Hansen 1999 (healthy humans): GH-specific release without significant cortisol or prolactin rise. Small Phase 2 in post-surgical patients (Goetze 2009): faster GI recovery reported. Body-composition effects usually reported after 4–6 wks; no long-term adult data.

SAFETY Generally well-tolerated short-term · GHSR agonism increases appetite · FDA 503A Category 2 bulk substance (compounding restricted, 2023)

ADVERSE Generally well tolerated short term: mild headache, transient flushing, increased appetite (GHSR agonism), water retention, injection-site reactions. Long-term effects uncharacterized.

CLINICAL CAVEAT GH release is modest alone and usually paired with a GHRH analogue (CJC-1295 no-DAC or sermorelin) for combined GHRH + GHSR signaling. The oral small-molecule MK-677 (ibutamoren) is functionally analogous but not a peptide and remains investigational. No adult anti-aging RCTs.

KEY REFERENCES

- Raun K et al. Eur J Endocrinol 1998;139:552 (PMID 9849822)
- Hansen BS et al. Eur J Endocrinol 1999;141:180
- Goetze O et al. Growth Horm IGF Res 2009;19:116

Research only · gray-market

GRADE D

GH SECRETAGOGUE (GHSR AGONIST)

Hexarelin

Examorelin · research only

MECHANISM

GHS-R1a (ghrelin receptor) agonist — more potent acute GH release than ipamorelin or GHRP-6, but less selective. Also activates cardiac GHS-R1a/CD36; preclinical and small human studies suggest GH-independent cardioprotective signaling. A single SC 1–2 µg/kg can drive GH peaks of 30–80 ng/mL within 15–30 min, exceeding GHRH alone.

INDICATIONS

- None approved
- Investigational: GH deficiency, heart failure
- Off-label/gray-market: GH optimization, recovery

DOSING · reported-use only, not recommended dosing; no approved indication

Research/off-label 100–200 µg SC 1–3×/day; empty stomach; cycle
≈4 weeks on, 2–4 weeks off

Clinical trial range 1–2 µg/kg SC daily; can combine with Mod GRF
1–29

EVIDENCE Single doses produce very high acute GH peaks (30–80 ng/mL). Small studies suggest CD36-mediated cardioprotection — improved cardiac function in heart failure and protection from ischemia-reperfusion. Phase 2 completed for GHD and heart failure; no Phase 3, no approval. GH response desensitizes within 2–4 wks of daily use.

SAFETY Cortisol and prolactin elevation (more than ipamorelin) · Rapid desensitization (2–4 wks of daily use) · No long-term data · Not approved anywhere

ADVERSE Increased ACTH/cortisol and prolactin (more HPA activation than ipamorelin), water retention, fatigue, sleep changes. GH response wanes within 2–4 wks of daily dosing (tachyphylaxis).

CLINICAL CAVEAT Among the most potent GHRPs for acute GH release, but rapid desensitization forces cycling and it raises cortisol/prolactin more than ipamorelin — so ipamorelin is preferred for most GH-optimization use. Preclinical and small human studies suggest CD36-mediated cardioprotection. FDA 503A Category 2 (compounding-restricted).

KEY REFERENCES

- Imbimbo BP et al. JCEM 1994;78:693
- Ghigo E et al. JCEM 1994;79:1407

Research only · gray-market · no human therapeutic approval

GRADE C

GH SECRETAGOGUE (GHSR AGONIST)

GHRP-2

Pralmorelin · GHRP Kaken (Japan, diagnostic)

MECHANISM

GHS-R1a (ghrelin receptor) agonist. More potent GH release than GHRP-6 with somewhat less appetite stimulation, but raises cortisol and prolactin more than ipamorelin.

INDICATIONS

- Japan: GH-deficiency diagnostic (pralmorelin)
- Off-label/gray-market: GH/IGF-1 optimization, body composition

DOSING

Diagnostic (Japan)	100 µg IV bolus, single dose for GH-stimulation testing
Off-label SC	100–300 µg SC 1–3×/day; empty stomach, cycle to limit desensitization · <i>not FDA-validated</i>

EVIDENCE GH-stimulation pharmacology well-characterized in humans; pralmorelin is validated as a Japanese GHD diagnostic. No Phase 3 therapeutic trials or Western approval; off-label optimization use rests on anecdote.

SAFETY Appetite increase (less than GHRP-6) · Cortisol and prolactin elevation · Water retention · FDA 503A Category 2 (compounding-restricted)

ADVERSE Increased appetite (GHSR agonism, milder than GHRP-6), transient cortisol and prolactin rise, water retention, injection-site reactions. Long-term therapeutic safety uncharacterized.

CLINICAL CAVEAT The only GHRP with a regulatory approval — pralmorelin is a licensed GH-stimulation diagnostic in Japan, not a therapy. Off-label optimization use is unstudied and raises cortisol/prolactin more than ipamorelin, which is preferred for clean GH release. FDA 503A Category 2 (compounding-restricted).

KEY REFERENCES

- Bowers CY. J Clin Endocrinol Metab 1994
- Pralmorelin (GHRP-2) — approved GH-stimulation diagnostic (Japan)

Approved in Japan (diagnostic) · research/gray-market elsewhere

GRADE D

GH SECRETAGOGUE (GHSR AGONIST)

GHRP-6

Research only · compounded

MECHANISM

GHS-R1a (ghrelin receptor) agonist. Potent GH release, but associated with marked appetite stimulation, and it raises cortisol, prolactin, and ACTH more than ipamorelin.

INDICATIONS

- None approved
- Off-label/gray-market: GH optimization, appetite stimulation, recovery

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label SC 100–300 µg SC 1–3×/day; empty stomach, cycle to limit desensitization

Stacking Often paired with a GHRH analogue (Mod GRF / CJC-1295)

EVIDENCE Historically pivotal in characterizing the ghrelin/GHS-R axis; demonstrated GH release in human studies. No Phase 3 efficacy or approval; off-label use rests on anecdote and the marked appetite effect.

SAFETY Marked appetite stimulation · Cortisol, prolactin, and ACTH elevation · Water retention · FDA 503A Category 2 (compounding-restricted)

ADVERSE Pronounced hunger (strongest of the GHRPs), transient cortisol/prolactin/ACTH rise, water retention, injection-site reactions, tingling. Long-term effects uncharacterized.

CLINICAL CAVEAT Mainly of historical and research interest; its strong appetite stimulation and HPA/prolactin activation make it less suitable than ipamorelin for GH optimization, though it is sometimes used deliberately as an appetite stimulant. Usually paired with a GHRH analogue. FDA 503A Category 2 (compounding-restricted).

KEY REFERENCES

- Bowers CY et al. Endocrinology 1984
- Bowers CY et al. J Clin Endocrinol Metab 1990

Research only · gray-market

GRADE C

GH FRAGMENT (177-191)

AOD-9604

Research only

MECHANISM

Stimulates lipolysis and inhibits lipogenesis via the $\beta 3$ -adrenergic-receptor pathway, mimicking the fat-metabolizing action of HGH but without affecting IGF-1, blood glucose, or growth.

INDICATIONS

- None approved for weight loss
- Marketed in some jurisdictions as a food/cosmetic ingredient (e.g., Australia, 2007)
- Off-label/compounded: body-composition and fat loss

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label SC (maintenance) 250–300 μg SC daily, morning and fasting, into abdominal fat

Loading 300–500 μg SC daily for the first 2–4 weeks, then reduce to maintenance

EVIDENCE Phase 2b (Stier 2007, n=536, 12 wks, oral formulation) failed to beat placebo for weight loss and development was terminated — though the oral route may have limited bioavailability vs SC. Preclinical and anecdotal SC use supports body-composition claims, but no positive RCT exists.

SAFETY Excellent safety across trials — AE rates similar to placebo · No serious AEs or metabolic perturbations · Marketed in Australia as a food ingredient (TGA, 2007); not an approved drug

ADVERSE Excellent tolerability — Phase 2b adverse-event rates matched placebo, with no serious events or metabolic perturbations. Unlike full-length HGH, it does not cause water retention, joint pain, or carpal-tunnel symptoms. Long-term data beyond trials are lacking.

CLINICAL CAVEAT No positive efficacy RCT — the Phase 2b trial failed its primary endpoint — yet AOD-9604 has an excellent safety record and remains widely compounded for body composition; best paired with a caloric deficit, resistance training, and protein over an 8–12 wk course. Marketed in Australia as a food ingredient; not FDA-approved for weight loss.

KEY REFERENCES

- Stier H et al. 2007 (Metabolic Pharmaceuticals) Phase 2b obesity trial, n=536
- Heffernan MA et al. Endocrinology 2001;142:5182

Not approved as a therapeutic · marketed in some jurisdictions as a food/cosmetic ingredient · compounded/research elsewhere · not FDA-approved

GRADE D

IGF-1 RECEPTOR AGONIST · SYNTHETIC GF ANALOGUE

IGF-1 LR3

Research only · gray-market (various manufacturers)

MECHANISM

Binds IGF-1R (a receptor tyrosine kinase), driving PI3K/AKT/mTOR and MAPK/ERK signaling — muscle protein synthesis, satellite-cell activation, glucose uptake, nitrogen retention, proliferation, and fat oxidation — independently of growth hormone. Reduced IGFBP binding gives ~2–3× the potency of native IGF-1.

INDICATIONS

- None approved
- Off-label gray-market: muscle hypertrophy, post-exercise recovery
- Tissue repair (preclinical) · athletic performance (no validated protocol)

DOSING · reported-use only, not recommended dosing; no approved indication

Gray-market SC/IM (bodybuilding)	20–100 µg post-workout or at target muscle
Cycle	4–6 weeks; doses extrapolated from community practice — not validated

EVIDENCE No human RCTs for body-composition or performance. Animal models show IGF-1R-mediated muscle hypertrophy, satellite-cell activation, and tendon repair; cell-culture data robust. No Phase 1/2 trials.

SAFETY Hypoglycemia — chief acute risk (insulin-receptor cross-reactivity) · Acromegalic-pattern soft-tissue enlargement with prolonged use · Theoretical oncogenic risk (IGF-1R overexpressed in cancers) · No human safety trials

ADVERSE Gray-market use: hypoglycemia (carry fast-acting glucose, monitor), joint/jaw pain, facial puffiness, water retention. Prolonged use may cause acromegaly-like soft-tissue enlargement. Theoretical tumor-promotion risk.

CLINICAL CAVEAT GRADE D — animal/preclinical and in-vitro only; no human efficacy trials, and healthy-aging claims are speculative. Gray-market purity is unverified and counterfeiting common (some products contain insulin). Chief acute risk: hypoglycemia from insulin-receptor cross-reactivity; long-term IGF-1R stimulation carries an unquantifiable oncogenic risk. Not licensed for human use.

KEY REFERENCES

- Francis GL et al. J Mol Endocrinol 1992;8:213
- Ballard FJ et al. J Biol Chem 1997;272:21 (PMID 8995300)
- Tomas FM et al. Biochem J 1993;291:781

Research only · gray-market · not approved for human use in any jurisdiction

THERAPEUTIC CLASS

Regenerative Peptides

“The gap between compelling preclinical data and human proof of efficacy is nowhere more pronounced than in regenerative peptide medicine.”

7 ENTRIES

GRADE C

BODY PROTECTION COMPOUND PEPTIDE

BPC-157

Bepecin · BPC 157 Stable · compounded

MECHANISM

Pleiotropic; no identified cognate receptor. Reported actions: VEGFR2 upregulation, eNOS activation, FAK-paxillin modulation, and GH-receptor upregulation in fibroblasts and tendon.

INDICATIONS

- None approved
- Off-label: tendon/ligament injury, gut-barrier repair (IBD, ulcers, SIBO), post-op recovery

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label SC/IM 250–500 µg 1–2×/day — systemic plus gut healing

Off-label oral (gut) 500–1000 µg 1–2×/day, empty stomach ~30 min before meals — local mucosal contact

Topical cream (burns) 1–2% cream/gel 2×/day; superficial burns only

EVIDENCE Extensive rodent literature (35+ studies, largely the Sikiric/Zagreb group). Pliva Phase II trials (PL-10, PLD-116, PL14736) suggested IBD/fistula benefit but were never fully published; the only human Phase I PK study (NCT02637284, 2015) is also unpublished. Vasireddi 2025 review found zero orthopedic RCTs.

SAFETY Short-term tolerability good in limited human use; long-term safety not characterized · WADA Section S0 (all routes, no TUE) · Topical cream: superficial burns only — avoid deep or 3rd-degree burns

ADVERSE Limited human data. Reported: mild injection-site reactions, transient fatigue, headache, hot flashes, GI upset, occasional insomnia. No serious AEs in animal or limited human use.

CLINICAL CAVEAT No high-quality human efficacy trials — clinical use rests on preclinical (rodent) data and anecdote. For gut use, oral suits local issues and SC adds systemic effect; often stacked with TB-500 and used alongside conventional IBD therapy. Oral is legal as a Canada NHP; WADA-banned.

KEY REFERENCES

- Sikiric P et al. (original characterisation, Zagreb 1993)
- Vasireddi N et al. 2025
- Sikiric P et al. Curr Pharm Des 2018;24:1865 (PMID 29804527)

503A Cat 2 (US) · not approved · July 2026 PCAC review (UC)

GRADE D

THYMIC PEPTIDE (ACTIN-SEQUESTERING)

TB-500 (Thymosin β 4)

Research only · compounded

MECHANISM

Thymosin β 4 is the most abundant β -thymosin and a major intracellular peptide; it sequesters monomeric G-actin to regulate cytoskeletal dynamics, driving cell migration, wound healing, and tissue repair. Also anti-inflammatory, anti-fibrotic, and cytoprotective in preclinical models.

INDICATIONS

- None approved
- Off-label/functional medicine: soft-tissue injury recovery, chronic inflammation, tissue repair
- Veterinary (equine)

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label SC (loading) 2.5 mg 2 $\times$ /week for 4–6 weeks

Maintenance 2.5 mg weekly (or every 2 weeks); inject near injury or in abdominal fat

Stack with BPC-157 Commonly co-administered for tissue repair

EVIDENCE RegeneRx RGN-259 (ophthalmic T β 4) completed Phase 3 for neurotrophic keratitis; systemic human data remain limited. Rodent models show dermal wound healing, cardiac repair after MI, reduced corneal inflammation, and hair-follicle stem-cell migration. Injury-healing effects may take 2–4 wks.

SAFETY Generally well-tolerated · Long-term safety unknown · WADA-prohibited since 2010 · Compounding restrictions in multiple jurisdictions

ADVERSE Limited safety data in humans. Reported by users: injection-site reactions, transient fatigue, head rush, lightheadedness, lethargy. Animal studies have not flagged major toxicity.

CLINICAL CAVEAT TB-500 — sold as a fragment/analog of thymosin β 4 — has no human efficacy data of its own; gray-market use rests on anecdote and animal repair models, and loading-then-maintenance protocols are anecdotal. Often stacked with BPC-157. WADA-prohibited.

KEY REFERENCES

- Goldstein AL et al. Ann N Y Acad Sci 2010;1194:1
- Crockford D et al. Ann N Y Acad Sci 2010;1194:179

Research only · gray-market · WADA-prohibited · July 2026 PCAC review (wound healing)

GRADE B

THYMIC PEPTIDE (IMMUNOMODULATOR)

Thymosin α 1

Zadaxin · Thymalfasin

MECHANISM

Modulates T-cell and dendritic-cell function; enhances Th1 responses via TLR9 signaling on plasmacytoid dendritic cells.

INDICATIONS

- Hepatitis B/C (in approved countries)
- Immunomodulation adjunct to vaccines
- Cancer adjunct in some protocols

DOSING · *reported-use only, not recommended dosing; no approved indication*

SC 2×/week

1.6 mg per dose, typically 6+ months

EVIDENCE Multiple RCTs for chronic hepatitis B/C show modest virologic response improvement. Sepsis trial (CONSIDER) failed. Tumor adjunct evidence is mixed.

SAFETY Well-tolerated; injection site reactions most common · Autoimmune disease theoretical concern · Pregnancy category not established

ADVERSE Generally well tolerated. Most common: injection-site reactions (transient), mild fatigue, chills. Transient elevation in autoantibody titers reported. No major safety signals in 30+ years of clinical use in approved markets.

CLINICAL CAVEAT Holds FDA orphan-drug designations (hepatocellular carcinoma, melanoma IIb–IV, DiGeorge) but no NDA marketing approval — different regulatory categories that are commonly conflated. Hepatitis B/C use is now obsolete (replaced by DAAs and nucleos(t)ides); current interest is sepsis, oncology adjunct, and immunocompromised vaccine response. Phase 3 complete for hepatitis and sepsis; approved in 35+ countries, no US registration.

KEY REFERENCES

- Camps E et al. — hepatitis B trials (multiple, foreign-approved)
- Wu J et al. Crit Care 2013;17:R8
- Garaci E et al. Ann N Y Acad Sci 2007;1112:14

Marketed in 35+ countries · FDA orphan drug designations only · removed from FDA 503A Category 2 bulk substances list in Sept 2024 (PCAC review pending)

GRADE D

ZINC-DEPENDENT THYMIC NONAPEPTIDE

Thymulin

Facteur Thymique Sérique (FTS) · research compound

MECHANISM

Active (zinc-bound) thymulin binds receptors on T-lymphocytes and NK cells, promoting T-cell differentiation and modulating cytokine balance (IL-2, IFN- γ). Serum levels decline with age and with thymic involution; zinc deficiency reduces the active form.

INDICATIONS

- None approved
- Investigational: age-related immune decline, post-chemo immune reconstitution; off-label gray-market: marketed for longevity claims

DOSING · reported-use only, not recommended dosing; no approved indication

SC (standard)	1–2 mg 2–3 $\times$ /week — immune optimization
SC (intensive)	2–3 mg 3–4 $\times$ /week — post-illness recovery
Intranasal	500–1000 μ g daily — non-invasive option

EVIDENCE Decades of preclinical work characterizing the zinc-thymulin axis (Bach, Dardenne, Goldstein groups). Limited small clinical work in immunology — older European studies in primary immunodeficiency and aging. No modern RCTs for current marketed indications.

SAFETY No modern human safety data · Generally well-tolerated in historical clinical use · Long-term effects unknown

ADVERSE Reported in historical clinical use: well tolerated, occasional injection-site reactions. No major safety signals at physiologic-range dosing. Long-term immune-modulation effects uncharacterized.

CLINICAL CAVEAT Active form requires zinc — co-supplement 15–30 mg elemental zinc daily for activation. Functional-medicine protocols run 8–12 wks (evening dosing; effects in ~2–4 wks), then reassess immune markers; often cycled 12 wks on / 4 off and combined with thymosin α 1. Evidence base: extensive preclinical work (Bach/Dardenne) plus small historical clinical studies — no modern RCTs.

KEY REFERENCES

- Bach JF, Dardenne M et al. *Nature* 1977;266:55
- Dardenne M et al. *PNAS* 1982;79:5370
- Reggiani PC et al. *Curr Pharm Des* 2014;20:4690

Research only · gray-market · unapproved

GRADE C

A-MSH C-TERMINAL TRIPEPTIDE

KPV

Research only · gray-market

MECHANISM

Anti-inflammatory; modulates NF- κ B signaling and inflammatory cytokine production without binding MC receptors. Specific molecular target remains uncharacterized.

INDICATIONS

- None approved
- Off-label/gray-market: inflammatory conditions, IBD, skin inflammation, post-procedure healing

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label SC 200–500 μ g daily

Off-label oral 1–5 mg daily (absorption limited)

Common stack Combined with BPC-157 for tissue-healing protocols

EVIDENCE Preclinical data for inflammatory-bowel-disease models; small open-label studies for atopic dermatitis. No randomized human trials.

SAFETY Generally well-tolerated · Injection-site reactions possible · Long-term safety not characterized

ADVERSE Limited human data. Generally well tolerated; injection-site reactions reported. No serious AEs documented in animal or limited human use.

CLINICAL CAVEAT Evidence base: animal IBD models, small dermatology case reports, and anecdotal use — no human RCTs. Frequently sold in the 'KLOW stack' (KPV + GHK-Cu + BPC-157 + TB-500); synergy is plausible but untested. Not to be confused with full α -MSH (which has melanocortin-receptor activity). Practice notes: compounded into topical skincare for atopic dermatoses; a small intradermal injection at a painful GHK-Cu site can blunt local inflammation — anecdotal but common in peptide clinics.

KEY REFERENCES

- Kannengiesser K et al. *Inflamm Bowel Dis* 2008;14:324
- Dalmasso G et al. *Gastroenterology* 2008;134:166

Research only · gray-market · July 2026 PCAC review (wound healing, inflammatory conditions)

Follistatin

Research only (recombinant) · investigational gene therapy

MECHANISM

Binds and neutralizes activin A and myostatin (GDF-8). Myostatin is the principal negative regulator of muscle growth; follistatin's myostatin inhibition removes that brake, driving muscle hypertrophy.

INDICATIONS

- None approved
- Investigational: Duchenne muscular dystrophy (AAV-mediated gene therapy)
- Off-label gray-market: muscle hypertrophy, athletic enhancement

DOSING · reported-use only, not recommended dosing; no approved indication

Gray-market recombinant	100–200 µg SC every other day for 10–30 day cycles
AAV-FST gene therapy	One-time intramuscular delivery (investigational)

EVIDENCE Strong animal data for muscle hypertrophy with viral-vector delivery. AAV-FST gene therapy in Phase 1/2 for DMD. Recombinant protein has not advanced clinically and gray-market product purity is highly variable.

SAFETY Theoretical cardiac hypertrophy concerns (myostatin loss in animal models) · Tendon–muscle imbalance · Long-term immunogenicity unknown

ADVERSE Limited human data outside DMD gene-therapy trials. Reported in gray-market use: injection-site reactions, mild flushing. Theoretical concerns: cardiac hypertrophy (myostatin loss in animal models), tendon–muscle imbalance.

CLINICAL CAVEAT Evidence base: investigational gene-therapy Phase 1/2 (FS-344) for Duchenne; the injectable peptide form has no published human efficacy data. Two very different routes: AAV gene therapy (investigational, sustained expression) vs gray-market recombinant peptide injections (very short half-life, dubious efficacy). The latter is widely used in bodybuilding with little evidence of meaningful effect.

KEY REFERENCES

- Mendell JR et al. Mol Ther 2017;25:870 (PMID 28279843)
- Kota J et al. Sci Transl Med 2009;1:6ra15

Research only · gray-market (recombinant) · Phase 1/2 for AAV-FST gene therapy in DMD

GRADE B

NON-ERYTHROPOIETIC EPO ANALOG

ARA-290

Cibinetide · Araim Pharmaceuticals (investigational)

MECHANISM

Activates the innate-repair receptor (IRR), a heterodimer of EPOR and CD131 (BCR), expressed on damaged tissue (neurons, endothelium, cardiomyocytes) but not on erythroid progenitors. Provides EPO's tissue-protective and anti-inflammatory effects without raising hemoglobin or risk of thromboembolism.

INDICATIONS

- None approved
- Investigational: sarcoid small-fiber neuropathy (Phase 2 positive)
- Investigational: diabetic peripheral neuropathy, neuropathic pain syndromes

DOSING · reported-use only, not recommended dosing; no approved indication

Phase 2 protocols (SC) 1–8 mg SC daily for 28 days; 4 mg commonly cited

EVIDENCE Phase 2 in sarcoidosis small-fiber neuropathy (Heij et al., Mol Med 2018) showed significant reduction in spontaneous pain vs placebo. Mechanism is well-characterized at the receptor level. Clinical development continues at Araim Pharmaceuticals; not yet Phase 3.

SAFETY Well-tolerated in Phase 2 · No erythropoietic effects observed (hemoglobin unchanged) · Long-term safety still pending Phase 3 data

ADVERSE Generally well tolerated. Mild injection-site reactions and headache reported. No erythropoietic effects detected; no thrombotic events in trial corpus to date.

CLINICAL CAVEAT ARA-290 is meaningfully distinct from the gray-market peptides on this list — it has a coherent receptor pharmacology, IND-cleared trials, and a defined commercial sponsor (Arim). Not yet FDA-approved. The non-erythropoietic separation of EPO's tissue-protective signaling is genuinely interesting drug discovery. Trial stage: Phase 2 positive in sarcoid small-fiber neuropathy (Heij 2018); Phase 3 pending.

KEY REFERENCES

- Heij L et al. Mol Med 2012;18:1430 (PMID 23168581)
- Dahan A et al. Mol Med 2013;19:334

Investigational (Phase 2) · Araim Pharmaceuticals · not approved

THERAPEUTIC CLASS

Dermatology & Cosmetic

“Copper, collagen, and tripeptides: the intersection of chemistry and aesthetics where evidence remains thin but patient demand is substantial.”

8 ENTRIES

GRADE C

COPPER TRIPEPTIDE

GHK-Cu

*Cosmetic ingredient · research compound***MECHANISM**

Copper-binding tripeptide; modulates copper delivery to enzymes including superoxide dismutase and lysyl oxidase. Activates wound-healing gene expression programs via copper-dependent signaling.

INDICATIONS

- Cosmetic: skin appearance, hair
- No drug approvals

DOSING · *reported-use only, not recommended dosing; no approved indication*

Topical cosmetic	0.05–2% formulations
Topical (microneedling)	0.5–1 mL of 0.01% solution per session
Off-label injectable	1–2 mg SC, 5–7 days per week

EVIDENCE Decades of dermatologic research (Pickart) support roles in collagen synthesis, antioxidant defense, and skin/hair benefits. Small cosmetic trials show modest skin-appearance benefit; no robust drug-indication RCTs.

SAFETY Well-tolerated topically · Injection-site pain and skin rash common with SC use · Hypersensitivity reactions reported but uncommon · Skin irritation possible at high concentrations · Chronic SC use: copper loading / zinc antagonism (GHK-Cu is a copper carrier) — monitor serum copper/zinc, supplement zinc if deficient, hold for hypercupremia.

ADVERSE Topical: mild erythema, dryness, irritation; contact dermatitis in sensitized users. Injectable: injection-site pain, erythema, induration, local rash. Hypersensitivity (urticaria, angioedema) uncommon but reported.

CLINICAL CAVEAT Topical cosmetic RCTs and preclinical wound-healing models provide some evidence, though efficacy claims often exceed the data; injectable human use rests on anecdote, no RCTs. The only peptide here with a recognized US compounding pathway (FDA 503A Category 1, non-injectable). Often sold as the “GLOW stack” (GHK-Cu + BPC-157 + TB-500); no RCT for the combination.

KEY REFERENCES

- Pickart L. *Nature* 1973;243:85
- Pickart L, Margolina A. *Int J Mol Sci* 2018;19:1987 (PMID 29986520)
- Leyden J et al. (multiple cosmetic RCTs, 2002 onward, manufacturer-sponsored)

FDA 503A Cat 1 (non-injectable, US) · Canada NHPID · 503A Cat 2 (injectable)

GRADE C

COPPER-FREE TRIPEPTIDE

GHK

Research compound · cosmetic ingredient

MECHANISM

Same fundamental biology as GHK-Cu — binds endogenous copper from albumin or extracellular pools to deliver copper to enzymes like superoxide dismutase and lysyl oxidase. Activates wound-healing and antioxidant gene-expression programs.

INDICATIONS

- None approved
- Cosmetic: skin appearance, hair
- Off-label/investigational: oncology adjunct claims (limited evidence)

DOSING · *reported-use only, not recommended dosing; no approved indication*

Off-label SC 2.5 mg SC three times weekly (gray-market oncology protocol)

Cosmetic topical 0.1–2% in serum or cream formulations

EVIDENCE Same in-vitro literature as GHK-Cu (Pickart et al., 2015, 2017). Whether the uncomplexed and copper-complexed forms produce clinically distinct outcomes in vivo is unresolved — they likely converge on the same downstream biology.

SAFETY Generally well tolerated topically · Injection-site reactions similar to but milder than GHK-Cu · No copper-overload concern (no pre-complexed copper)

ADVERSE Mild injection-site irritation; occasional rash. Reportedly better-tolerated SC than GHK-Cu because the copper is not pre-loaded onto the peptide.

CLINICAL CAVEAT Evidence base: limited direct human data — mechanism inferred from the GHK-Cu literature and topical cosmetic studies; injectable use rests on anecdote and the same in-vitro work as GHK-Cu, with no human RCTs. Sometimes marketed as distinct from GHK-Cu, but the two interconvert in tissue depending on local copper; the clearest practical difference is injection-site tolerability (GHK is gentler). Oncology claims rest on thin evidence and should not displace evidence-based cancer therapy.

KEY REFERENCES

- Pickart L. Nature 1973;243:85
- Pickart L et al. Biomed Res Int 2015;648108

Research only · not regulated as a drug

GRADE D

COSMETIC MATRIKINE LIPOPEPTIDE

Palmitoyl Tripeptide-1

Pal-GHK · cosmetic ingredient

MECHANISM

Matrikine that signals dermal fibroblasts to upregulate collagen (types I, III, IV), fibronectin, and glycosaminoglycans, mimicking ECM-fragment repair signaling. Palmitoylation enhances skin penetration. Topical.

INDICATIONS

- None (cosmetic ingredient, not a drug)
- Topical: anti-wrinkle, skin firmness, photoaging

DOSING · *reported-use only, not recommended dosing; no approved indication*

Topical Trace concentrations ($\approx$ 1–10 ppm) in cosmetic serums/creams, 1–2 $\times$ /day

EVIDENCE Cosmetic in-vitro and small clinical studies report increased collagen synthesis and modest wrinkle improvement; no independent drug-level RCTs.

SAFETY Topical use generally well tolerated · Rare mild irritation or sensitivity · Cosmetic ingredient — no drug oversight

ADVERSE Generally well tolerated topically; rare mild irritation or contact sensitivity. No systemic effects expected at cosmetic concentrations.

CLINICAL CAVEAT The palmitoylated GHK matrikine at the core of Matrixyl-type blends; cosmetic, not a drug. Used at very low (ppm) concentrations; efficacy is modest and formulation-dependent, supported by cosmetic studies rather than independent RCTs. Topical only — no injectable use.

KEY REFERENCES

- Lintner K, Peschard O. Int J Cosmet Sci 2000;22:207
- Sederma technical literature — pal-GHK collagen-stimulation data (manufacturer)

Cosmetic ingredient · not a drug · topical only

GRADE D

COSMETIC MATRIKINE PEPTIDE BLEND

Matrixyl 3000

Matrixyl 3000 (Sederma) · cosmetic

MECHANISM

Matrikines act as ECM-fragment signals: pal-GHK and pal-GQPR stimulate fibroblast synthesis of collagen (I, III), fibronectin, hyaluronic acid, and glycosaminoglycans, and damp IL-6-type inflammation. Palmitoylation aids penetration. Topical.

INDICATIONS

- None (cosmetic ingredient, not a drug)
- Topical: wrinkle reduction, skin firmness, anti-aging

DOSING · reported-use only, not recommended dosing; no approved indication

Topical 2–8% of the trade-name blend in serum/cream,
1–2×/day

EVIDENCE Manufacturer and small cosmetic studies report reduced wrinkle depth and volume over 1–2 months vs placebo; independent peer-reviewed RCTs are limited. No drug-level efficacy data.

SAFETY Topical use generally well tolerated · Mild irritation or contact dermatitis possible
· Cosmetic ingredient — no drug oversight

ADVERSE Topical use is generally well tolerated; occasional mild irritation, redness, or contact sensitivity. No systemic exposure expected from cosmetic concentrations.

CLINICAL CAVEAT A cosmetic matrikine blend (pal-GHK + pal-GQPR), not a drug. Topical only; effect size is modest and formulation-dependent.

KEY REFERENCES

- Sederma (Croda) technical dossier — Matrixyl 3000 (pal-GHK + pal-GQPR) cosmetic studies
- Lintner K. Cosmet Toilet 2007

Cosmetic ingredient · not a drug · topical only

GRADE D

CELL-PENETRATING WNT-PATHWAY ACTIVATOR

PTD-DBM

Topical hair-growth research compound

MECHANISM

Penetrates hair-follicle cells via the TAT-PTD domain. Inside the cell, the DBM peptide disrupts CXXC5-Dishevelled binding, releasing Dvl to activate Wnt/ β -catenin signaling and induce the anagen growth phase in dermal papilla.

INDICATIONS

- None approved
- Off-label/compounded: androgenetic alopecia, post-chemotherapy alopecia (topical)

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label topical

0.1–0.5% concentration in spray vehicle, applied to scalp 1–2× daily; often combined with valproic acid 5% to enhance Wnt activation

EVIDENCE Preclinical and limited dermatology research from the Yonsei group (Choi et al., 2017). No large RCT. Used in compounding pharmacies, typically with valproic acid (a GSK-3 β inhibitor).

SAFETY Topical route limits systemic exposure · Scalp irritation possible · Long-term effects unknown

ADVERSE Scalp irritation, mild redness, itching. Rarely allergic contact dermatitis. Hair-shedding phase during early use as anagen induction begins (similar to minoxidil).

CLINICAL CAVEAT Mechanistically among the more interesting investigational hair-loss compounds on this list, but human efficacy data is thin — most use is via compounding pharmacies. Effective only when combined with a Wnt-pathway modulator (typically valproic acid). Preclinical only; no human RCTs.

KEY REFERENCES

- Choi S et al. (Yonsei lab) — Wnt-pathway PTD-DBM preclinical, mid-2010s

Research only · used in compounding pharmacies for hair growth

GRADE D

ZINC-BOUND THYMIC PEPTIDE

Zinc Thymulin

Compounded · research compound

MECHANISM

Parenterally, modulates T-cell maturation and thymopoiesis. Topical hair-growth mechanism less defined — proposed to optimize follicular microenvironment and dermal-papilla signaling. The zinc cofactor may itself contribute (zinc deficiency is associated with diffuse alopecia).

INDICATIONS

- None approved
- Off-label topical: androgenetic alopecia (typically combined with GHK-Cu)

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label topical

0.2% in 30 mL topiclick or spray vehicle, 1 mL applied to scalp daily

EVIDENCE Preclinical and small case-series for hair growth. No published RCT. Most clinical use is in compounded combination products with GHK-Cu and other adjuncts; isolating zinc thymulin's contribution is difficult.

SAFETY Topical route limits systemic exposure · Scalp irritation occasional · No long-term safety data

ADVERSE Mild scalp irritation, occasional hyperpigmentation. No serious adverse events documented in case-series use.

CLINICAL CAVEAT Reasonable adjunct in combination topical hair products where each component is well-tolerated; not a stand-alone therapy with proven efficacy. The role of the zinc cofactor itself complicates the question of whether 'zinc thymulin' acts as a peptide or as a vehicle for topical zinc. Trial stage: preclinical and small case-series for hair; no RCTs.

KEY REFERENCES

- Bach JF et al. Nature 1977;266:55
- Caroleo MC et al. (zinc thymulin hair-loss case series, small)

Research only · compounded for cosmetic hair use

GRADE D

SNAP-25 N-TERMINAL MIMETIC ("TOPICAL BOTULINUM")

Argireline

Acetyl Hexapeptide-3 · Acetyl Hexapeptide-8 (INCI)

MECHANISM

Mimics the N-terminus of SNAP-25, competing with native SNAP-25 for incorporation into the SNARE complex. In theory, reduces vesicle fusion at the neuromuscular junction, attenuating muscle contraction and expression lines. Topical penetration to facial muscle is mechanistically debated.

INDICATIONS

- None approved (cosmetic ingredient only)
- Cosmetic: expression lines, fine wrinkles (forehead, peri-orbital)

DOSING · reported-use only, not recommended dosing; no approved indication

Cosmetic topical

5–10% in serum (best penetration), morning and evening

Application

Clean, dry skin before moisturizer; allow ~2–3 min to absorb; visible results in 4–6 weeks

EVIDENCE Multiple manufacturer-sponsored cosmetic studies showing modest improvement in wrinkle depth. Independent evidence less conclusive — transdermal penetration to facial muscle is implausible at topical concentrations; any cosmetic effect may reflect vehicle, hydration, or hygroscopic action of the peptide rather than SNARE inhibition.

SAFETY Cosmetic ingredient with two decades of broad use · Generally well-tolerated topically · Minimal systemic exposure

ADVERSE Rare contact dermatitis. No serious adverse events in cosmetic use. Theoretical concern about systemic SNARE inhibition is not borne out — topical bioavailability to neuromuscular junction is essentially zero.

CLINICAL CAVEAT Marketed as "topical botox" but the mechanism requires the peptide to reach the neuromuscular junction beneath facial muscles — most dermatology authorities consider this implausible at cosmetic concentrations. Any cosmetic effect is modest and not a substitute for botulinum injections. Pairs well with Matrixyl, GHK-Cu, and hyaluronic acid; store cool and dark to preserve peptide stability.

KEY REFERENCES

- Blanes-Mira C et al. Int J Cosmet Sci 2002;24:303
- Wang Y et al. (manufacturer follow-ups, multiple)

Cosmetic ingredient · not a drug

GRADE D

STABILIZED ENKEPHALIN PENTAPEPTIDE (COSMETIC)

Leuphasyl

Pentapeptide-18 (INCI) · enkephalin analog

MECHANISM

Binds enkephalin receptors (presumed δ -opioid receptors) at the neuromuscular junction, attenuating acetylcholine release and thereby reducing muscle contraction. Same caveat as Argireline regarding topical penetration to the neuromuscular junction.

INDICATIONS

- None approved (cosmetic ingredient only)
- Cosmetic: expression lines, fine wrinkles (typically combined with Argireline)

DOSING · reported-use only, not recommended dosing; no approved indication

Expression-line serum

Pentapeptide-18 3–5%, forehead and eye area, morning and evening

Combination

Pentapeptide-18 3% + Argireline 5% on expression lines, 1–2×/day — dual neurotransmitter inhibition

EVIDENCE Manufacturer-sponsored cosmetic studies report modest additional benefit when combined with Argireline. Independent evidence is limited.

SAFETY Cosmetic ingredient · Generally well-tolerated topically · Minimal systemic exposure

ADVERSE Local tolerability mirrors Argireline — rare contact dermatitis; no serious AEs in cosmetic use.

CLINICAL CAVEAT Like Argireline, the topical-botox premise depends on transdermal delivery to the neuromuscular junction, which is mechanistically implausible at cosmetic concentrations. The rationale for pairing with Argireline is dual-mechanism action — enkephalin-mediated reduction of acetylcholine release plus Argireline's SNARE interference; visible softening is reported over 4–6 wks. Effect is modest at best and synergistic claims rest on company-sponsored data. Store sealed, away from light.

KEY REFERENCES

- Lipotec (manufacturer) cosmetic studies — limited independent peer-reviewed evidence

Cosmetic ingredient · not a drug

THERAPEUTIC CLASS

Bone Metabolism

“Teriparatide and abaloparatide reversed a therapeutic paradigm — proving that activating the PTH receptor intermittently builds bone, while sustained activation destroys it.”

3 ENTRIES

GRADE A

PTH(1-34) ANALOGUE

Teriparatide

Forteo · Bonsity · biosimilars

MECHANISM

Intermittent PTH receptor (PTH1R) activation on osteoblasts; pulsed exposure favors bone formation over resorption (opposite of chronically elevated PTH).

INDICATIONS

- Severe osteoporosis (postmenopausal women, men, glucocorticoid-induced)
- Patients at high risk of fracture

DOSING

SC daily	20 µg daily
Max duration	2 years lifetime

EVIDENCE Pivotal Fracture Prevention Trial showed 65% reduction in vertebral fracture and 53% reduction in non-vertebral fracture vs placebo.

SAFETY Boxed warning for osteosarcoma (rat data; not seen in human postmarketing) · Hypercalcemia, hypercalciuria · Orthostatic hypotension · Contraindicated in prior radiation, Paget disease, bone malignancy

ADVERSE Most common: nausea (8%), headache, dizziness, leg cramps, arthralgia. Transient hypercalcemia and hypercalciuria. Orthostatic hypotension after first doses. Boxed warning for osteosarcoma (rat data; not clearly seen in human postmarketing).

CLINICAL CAVEAT Lifetime 24-month maximum due to rat osteosarcoma signal (not seen in 20+ years of human postmarketing). Sequencing teriparatide → bisphosphonate preserves the anabolic bone gains after discontinuation.

KEY REFERENCES

- Neer RM et al. Fracture Prevention Trial. NEJM 2001;344:1434 (PMID 11346807)
- Saag KG et al. (vs alendronate) NEJM 2007;357:2028 (PMID 18003959)

FDA-approved · EMA · all major regulators

GRADE A

PTHrP(1-34) ANALOGUE

Abaloparatide

Tymlos

MECHANISM

Selectively activates the RG (rapid, reversible) conformation of PTH1R; theoretically more anabolic than teriparatide with less hypercalcemia risk.

INDICATIONS

- Postmenopausal osteoporosis (high fracture risk)
- Men with osteoporosis at high fracture risk

DOSING

SC daily	80 µg daily
Max duration	2 years lifetime

EVIDENCE ACTIVE trial: 86% reduction in new vertebral fracture vs placebo over 18 mo. ACTIVEExtend follow-on with alendronate showed durable benefit.

SAFETY Boxed warning for osteosarcoma (class effect) · Hypercalcemia (less than teriparatide) · Orthostatic hypotension · Same contraindications as teriparatide

ADVERSE Most common: hypercalciuria (11%), dizziness (10%), nausea (8%), headache (8%), palpitations, fatigue, abdominal pain, vertigo. Less hypercalcemia than teriparatide. Same boxed osteosarcoma class warning.

CLINICAL CAVEAT PTHrP analogue with less hypercalcemia than teriparatide but the same 24-month lifetime limit and osteosarcoma class warning. Real-world adoption limited by cost and prior-authorization barriers.

KEY REFERENCES

- Miller PD et al. ACTIVE. JAMA 2016;316:722 (PMID 27536855)
- Cosman F et al. ACTIVEExtend. Mayo Clin Proc 2017;92:200 (PMID 28160873)

FDA-approved (Tymlos, 2017) · EMA (Eladynos, 2022)

GRADE A

CALCIUM-REGULATING PEPTIDE

Calcitonin (salmon)

Miacalcin · Fortical

MECHANISM

Inhibits osteoclast activity via the calcitonin receptor (a Gs-coupled GPCR). Also has central analgesic effects via spinal calcitonin receptors.

INDICATIONS

- Paget disease
- Hypercalcemia of malignancy
- Postmenopausal osteoporosis (second-line, intranasal use)
- Vertebral compression fracture pain

DOSING

Intranasal (osteoporosis)	200 IU one nostril daily, alternating
SC/IM (Paget)	100 IU daily, then maintenance
SC/IM (hypercalcemia)	4–8 IU/kg q6–12h

EVIDENCE Modest fracture-reduction effect in PROOF trial (~33% vertebral); no effect on hip or non-vertebral fractures. EMA restricted long-term use in 2012 due to malignancy signal.

SAFETY Possible increase in malignancy with long-term use (EMA review) · Nasal irritation · GI side effects with parenteral use · Allergic reactions (salmon protein)

ADVERSE Intranasal: rhinitis, nasal irritation, epistaxis. Parenteral: nausea, vomiting, flushing, injection-site reactions. Increased malignancy signal with chronic use prompted EMA restrictions (2012). Hypersensitivity to salmon protein possible.

CLINICAL CAVEAT Intranasal use was restricted by EMA in 2012 due to a malignancy signal in long-term users. Now primarily short-term for hypercalcemia, Paget disease, and vertebral compression fracture pain.

KEY REFERENCES

- Chesnut CH et al. PROOF. Am J Med 2000;109:267 (PMID 10996576)
- Cochrane review (Wells GA et al. 2008)

FDA-approved · EMA (restricted)

THERAPEUTIC CLASS

Somatostatin Analogues

“Somatostatin analogues are among the most evidence-rich peptide drugs in medicine, with decades of pivotal trial data across three receptor-selective compounds.”

3 ENTRIES

GRADE A

SOMATOSTATIN ANALOGUE

Octreotide

Sandostatin · Sandostatin LAR · Mycapssa

MECHANISM

Agonist at somatostatin receptors SSTR2 and SSTR5 (less so SSTR3); inhibits hormone secretion (GH, insulin, glucagon, gastrin) and reduces neuroendocrine tumor proliferation.

INDICATIONS

- Acromegaly
- Carcinoid syndrome
- VIPoma
- Gastrointestinal bleeding from varices
- Diagnostic (Octreoscan)

DOSING

SC 3×/day (acute)	100–600 µg/day in divided doses
IM monthly (LAR)	10–40 mg every 4 weeks
Oral (Mycapssa)	20 mg 2×/day, titrate to 40–80 mg/day

EVIDENCE Multiple RCTs and decades of clinical experience for acromegaly biochemical control (~50–60%) and carcinoid syndrome symptom relief.

SAFETY GI side effects (steatorrhea, nausea, gallstones) · Hypothyroidism · Hyperglycemia or hypoglycemia · Bradycardia

ADVERSE Most common: diarrhea (34%), abdominal cramps (28%), nausea (27%), steatorrhea. Cholelithiasis (gallstones) in 10–20% with chronic use. Hypothyroidism, glucose dysregulation (hyper- or hypoglycemia), bradycardia, injection-site pain.

CLINICAL CAVEAT Mycapssa (oral, 2020) is the first oral somatostatin analogue. Most patients still use IM LAR for convenience and durability; the radiolabeled Octreoscan form remains a standard neuroendocrine-tumor imaging agent.

KEY REFERENCES

- Rinke A et al. PROMID. JCO 2009;27:4656 (PMID 19704057)
- Caplin ME et al. CLARINET. NEJM 2014;371:224 (PMID 25014687)

FDA-approved · EMA · all major regulators

GRADE A

SOMATOSTATIN ANALOGUE

Lanreotide

Somatuline Depot

MECHANISM

Same as octreotide; SSTR2 and SSTR5 agonist, with depot formulation providing month-long exposure.

INDICATIONS

- Acromegaly
- Locally advanced or metastatic GEP-NET
- Carcinoid syndrome

DOSING

Deep SC monthly 120 mg every 4 weeks (typical maintenance)

EVIDENCE CLARINET trial: significantly prolonged progression-free survival in metastatic GEP-NET (HR 0.47). Equivalent efficacy to octreotide LAR for acromegaly.

SAFETY GI side effects · Gallbladder issues · Glucose dysregulation · Injection-site reactions

ADVERSE Most common: diarrhea (49%), cholelithiasis (27%), abdominal pain, flatulence, nausea, injection-site reactions. Hyperglycemia, bradycardia. Long-term gallbladder disease is a class effect of somatostatin analogues.

CLINICAL CAVEAT Equivalent efficacy to octreotide LAR for acromegaly; the pre-filled deep-SC depot syringe is easier for patients to self-administer than octreotide LAR.

KEY REFERENCES

- Caplin ME et al. CLARINET. NEJM 2014;371:224 (PMID 25014687)
- Wolin EM et al. ELECT. JCEM 2015;100:3825 (PMID 26244380)

FDA-approved (Somatuline Depot, 2007) · EMA

GRADE A

MULTI-RECEPTOR SOMATOSTATIN ANALOGUE

Pasireotide

Signifor · Signifor LAR

MECHANISM

Pan-SSTR agonist with highest affinity for SSTR5 (relevant to ACTH-secreting pituitary adenomas in Cushing disease).

INDICATIONS

- Cushing disease (when surgery has failed or is contraindicated)
- Acromegaly (octreotide/lanreotide-resistant)

DOSING

SC 2×/day (Cushing)	0.3–0.9 mg 2×/day (start 0.6 or 0.9 mg, titrate)
IM monthly (LAR)	20–60 mg every 4 weeks

EVIDENCE Phase 3 for Cushing: ~25% biochemical normalization. PAOLA trial in acromegaly showed superiority to octreotide/lanreotide in patients with inadequate response.

SAFETY Hyperglycemia is the principal limiting toxicity (much more common than with octreotide/lanreotide) · GI side effects · QT prolongation · Cholelithiasis

ADVERSE Most common: hyperglycemia (much more than octreotide/lanreotide — 73% in Cushing patients), diarrhea (58%), cholelithiasis (30%), nausea, hypocortisolism, QT prolongation. Hyperglycemia frequently requires anti-diabetic therapy.

CLINICAL CAVEAT Hyperglycemia is the dose-limiting toxicity, occurring in ~73% of Cushing patients — much more than octreotide or lanreotide. The pan-SSTR profile (especially SSTR5) explains both efficacy in Cushing and the metabolic adverse-event pattern.

KEY REFERENCES

- Colao A et al. PAOLA. Lancet Diabetes Endocrinol 2014;2:875 (PMID 25260838)
- Colao A et al. NEJM 2012;366:914 (PMID 22397653)

FDA-approved (Signifor, 2012) · EMA

THERAPEUTIC CLASS

Melanocortin & Sexual

“The melanocortin system governs pigmentation, sexual desire, energy balance, and inflammation — a single receptor family with remarkable biological reach.”

3 ENTRIES

Bremelanotide / PT-141

Vyleesi (SC) · PT-141 (*gray-market intranasal alias*)

MECHANISM

Melanocortin (MC4R) agonist; central MC4R activation drives sexual desire — acting on the brain (desire), not blood flow like PDE5 inhibitors (Viagra/Cialis), and can initiate arousal without direct stimulation. Intranasal absorption is far more variable than SC.

INDICATIONS

- Hypoactive sexual desire disorder (HSDD) in premenopausal women (Vyleesi · SC)
- Off-label intranasal (PT-141): ED or desire in either sex — no validated protocol

DOSING

SC as-needed (Vyleesi)	1.75 mg ~45 min before activity; max 1 dose/24 h, 8/month
Off-label men (SC)	4–6 mg studied in Phase 1/2 (not approved) · <i>not FDA-validated</i>
Intranasal (PT-141 gray-market)	1–5 mg ~45 min before; no validated protocol · <i>not FDA-validated</i>

EVIDENCE RECONNECT trials (n>1200, SC): modest improvement in desire and distress vs placebo (~25% vs 17% responders). The Palatin intranasal program halted at Phase 2 for BP safety; no published intranasal efficacy trials.

SAFETY Transient BP elevation (peaks 2–4h) — halted the intranasal program · Contraindicated in uncontrolled hypertension and CV disease · Intranasal: nasal irritation and unpredictable absorption

ADVERSE Vyleesi SC: nausea (40%, usually mild/transient — consider ondansetron pretreatment), flushing, injection-site reactions, headache. Transient BP rise peaks 2–4h. Focal hyperpigmentation with repeated use. Intranasal adds nasal irritation and variable absorption; priapism reported in men.

CLINICAL CAVEAT Default to bremelanotide (Vyleesi) SC; the gray-market intranasal “PT-141” is the same molecule with no validated dose and the BP signal that ended the nasal program. SC peaks ~4 h and clears in 8–10 h; effects last ~6–12 h. As-needed only; address pre-existing hypertension first. Male development was halted in 2007 over BP safety.

KEY REFERENCES

- Kingsberg SA et al. RECONNECT. *Obstet Gynecol* 2019;134:899 (PMID 31599840)
- Clayton AH et al. RECONNECT pooled. *J Sex Med* 2019;16:1953
- Diamond LE et al. *J Sex Med* 2004

FDA-approved 2019 (Vyleesi · SC · premenopausal HSDD) · intranasal PT-141: gray-market

GRADE C

NON-SELECTIVE MELANOCORTIN AGONIST

Melanotan II

Gray market only

MECHANISM

Synthetic cyclic heptapeptide α -MSH analogue (~1,000× more potent than native α -MSH); non-selective MC1R/MC3R/MC4R/MC5R agonist. MC1R drives melanogenesis (tanning); MC3R/MC4R drive the sexual effects. This scaffold led to the approved derivatives PT-141 and afamelanotide.

INDICATIONS

- None approved
- Off-label/gray-market: tanning, erectile dysfunction, sexual desire

DOSING · *reported-use only, not recommended dosing; no approved indication*

Conservative start

100–250 μ g SC every 2–3 days — assess tolerance, limit nausea

Tanning

Load 250–500 μ g SC daily $\times$ 10–14 d, then 1–2 $\times$ /week (with minimal UV)

Sexual (off-label)

ED ~0.025 mg/kg ($\approx$ 1.75–2 mg) or 0.5–1 mg for desire, 1–4 h before

EVIDENCE Phase 1 (Dorr 1996): 8/10 men achieved clinical erections ($\approx$ 38 vs 3 min on placebo, $p=0.0045$), with increased sexual desire in both sexes; no modern efficacy RCTs.

SAFETY Nausea, flushing, spontaneous erection / priapism · Cardiovascular concerns from rapid BP changes; melanoma/nevus cases reported

ADVERSE Common: nausea, vomiting, cramps, flushing, dizziness, fatigue, headache, anorexia, weight loss. Sexual: spontaneous arousal in both sexes; prolonged painful erections (priapism — emergency, can permanently damage tissue). Severe: new/darkening moles, melanoma (pigment changes can mask early skin-cancer signs); rhabdomyolysis; renal infarction and kidney dysfunction; hepatic injury; CNS effects (BBB-crossing) with anxiety, mood swings, brain swelling. No oversight of purity or dose.

CLINICAL CAVEAT A single Phase 1 (Dorr 1996) plus case reports of serious harm — no efficacy RCTs. Start low (100–250 μ g) and inject 2–3 h before bed on an empty stomach to limit nausea; tanning still needs some UV and is cycled 1–2 $\times$ /year. Unregulated injectable — regular nevus surveillance is essential.

KEY REFERENCES

- Dorr RT et al. Life Sci 1996;58:1777
- Case reports: melanoma, priapism, rhabdomyolysis (TGA/MHRA warnings)

Not approved · gray-market · purchase banned in several countries

GRADE A

MC1R AGONIST

Afamelanotide / Melanotan 1

Scenesse

MECHANISM

Selective MC1R agonist — unlike Melanotan II it does not activate MC3R/MC4R, so it lacks MT-II's sexual-arousal, nausea, and appetite effects (a key safety advantage). Drives eumelanin synthesis for photoprotection via UV absorption, antioxidant activity, DNA-repair enhancement, and anti-inflammatory effects.

INDICATIONS

— Erythropoietic protoporphyria (EPP)

DOSING

Implant

16 mg bioresorbable SC implant every ~2 months (3–4/year); placed by a trained clinician via a dedicated cannula

EVIDENCE Three pivotal Phase 3 trials (CUV029/030/039, 244 EPP adults). Langendonk 2015 (NEJM): significantly greater pain-free time in direct sunlight (≈64 h vs 41 h placebo). About 91% of real-world patients continue treatment; 10-year safety data show no new signals.

SAFETY Implant-site reactions · Headache, nausea · Skin/nevus pigmentation increase (expected) — dermatologic surveillance recommended

ADVERSE Most common: implant-site reactions (erythema, pain, hyperpigmentation — 28%), nausea, fatigue, headache, back pain, melanocytic naevus darkening, nasopharyngitis. Dermatologic surveillance recommended.

CLINICAL CAVEAT The approval covers only the Clinuvil 16 mg bioresorbable implant (Scenesse) every 2 mo for EPP — not generic afamelanotide powder, compounded forms, or research-grade Melanotan 1 vials. Conflating Scenesse approval with cosmetic-tanning melanotan products misrepresents the regulatory record.

KEY REFERENCES

- Langendonk JG et al. NEJM 2015;373:48 (PMID 26132941)
- Biolcati G et al. Br J Dermatol 2015;172:1601

FDA-approved as Scenesse (NDA 210797, Oct 2019) for EPP only · EMA-approved 2014 (exceptional circumstances)

THERAPEUTIC CLASS

Reproductive & Pituitary

“The posterior pituitary peptides were the first synthetic hormones to enter clinical medicine, and remain irreplaceable across obstetrics, fertility, and water homeostasis.”

6 ENTRIES

GRADE A

GNRH AGONIST

Leuprolide

Lupron · Eligard · Lupron Depot

MECHANISM

Initial GnRH receptor agonism causes LH/FSH surge ("flare"); sustained exposure desensitizes pituitary gonadotropes, → profound suppression of LH, FSH, testosterone, and estradiol.

INDICATIONS

- Advanced prostate cancer
- Endometriosis
- Uterine fibroids
- Central precocious puberty
- Gender-affirming care (puberty blocker)

DOSING

Daily SC	1 mg daily
Monthly depot IM/SC	7.5 mg every 4 weeks
3-month depot	22.5 mg
6-month depot	45 mg

EVIDENCE Standard of care for advanced prostate cancer for decades; survival benefit established in combination with antiandrogens.

SAFETY Initial testosterone surge can worsen disease (use with antiandrogen) · Hot flashes, sexual dysfunction, BMD loss, metabolic changes · Injection-site reactions

ADVERSE Most common: hot flashes (55%), decreased libido, erectile dysfunction, BMD loss with chronic use, weight gain, fatigue, mood changes, gynecomastia, injection-site reactions. Initial tumor flare from testosterone surge.

CLINICAL CAVEAT Initial testosterone surge ("flare") requires antiandrogen coverage for the first 2–4 wks in prostate cancer to prevent disease flare. Depot and daily formulations are clinically interchangeable.

KEY REFERENCES

- Crawford ED et al. NEJM 1989;321:419
- Lupron Depot trials (multiple) for endometriosis, central precocious puberty

FDA-approved · EMA · all major regulators

GRADE A

GnRH AGONIST

Goserelin

Zoladex

MECHANISM

Same mechanism as leuprolide; sustained GnRH receptor activation causing pituitary desensitization.

INDICATIONS

- Prostate cancer
- Breast cancer (premenopausal)
- Endometriosis
- Uterine fibroids

DOSING

Implant 3.6 mg every 28 days, or 10.8 mg every 12 weeks

EVIDENCE Equivalent efficacy to leuprolide for prostate cancer; preferred for breast cancer ovarian suppression based on SOFT/TEXT trials.

SAFETY Same as leuprolide (hot flashes, BMD loss, sexual dysfunction) · Implant site reactions · Tumor flare with initial dose

ADVERSE Hot flashes (62%), decreased libido, vaginal dryness (in women), BMD loss, mood changes, weight changes, headache, implant-site reactions. Tumor flare with initial dosing in prostate cancer.

CLINICAL CAVEAT Implant administration (vs leuprolide IM/SC) gives reliable depot release without reconstitution. SOFT and TEXT trials established it as the preferred GnRH agonist for premenopausal breast-cancer ovarian suppression.

KEY REFERENCES

- Goldhirsch A et al. ZIPP. Lancet Oncol 2001
- Klijn JG et al. JCO 2001;19:343

FDA-approved (Zoladex, 1989) · EMA · all major regulators

GRADE A

PREGNANCY-DERIVED GONADOTROPIN

hCG

Pregnyl · Novarel · Ovidrel (recombinant)

MECHANISM

Binds the LH/CG receptor (LHCGR) on Leydig cells (in men) and theca/corpus luteum cells (in women). In men, drives testosterone production. In women, supports corpus luteum and progesterone production.

INDICATIONS

- Cryptorchidism (pediatric)
- Hypogonadotropic hypogonadism (male)
- Ovulation induction (anovulatory infertility)
- IVF oocyte final maturation trigger

DOSING

Cryptorchidism	500–4000 IU IM weekly × 4–12 weeks
Male hypogonadism	500–2000 IU IM 2–3× weekly
IVF trigger	5000–10000 IU IM once (or recombinant Ovidrel 250 µg SC once)

EVIDENCE Decades of clinical use; well-established efficacy for ovulation induction and male hypogonadism. Strong evidence for IVF triggering (partially replaced by kisspeptin in OHSS-prone patients).

SAFETY Ovarian hyperstimulation syndrome (OHSS) is the principal serious risk · Thromboembolic events · Contraindicated in hormone-sensitive cancers

ADVERSE Most common: injection-site pain, fatigue, headache, mood changes, breast tenderness. Serious: ovarian hyperstimulation syndrome (OHSS — can be life-threatening), thromboembolic events, ovarian torsion. Gynecomastia possible in men with prolonged use.

CLINICAL CAVEAT Misused historically in the "HCG diet" (Simeons protocol) — repeatedly debunked; weight loss attributable to severe caloric restriction, not hCG. In men, used off-label to preserve testicular function during testosterone replacement therapy and to restore spermatogenesis post-cycle.

KEY REFERENCES

- Trimble EL et al. (older fertility-treatment literature)
- Brinholi FF et al. Front Endocrinol 2017

FDA-approved · all major regulators

GRADE B

HYPOTHALAMIC NEUROPEPTIDE

Kisspeptin-10

Investigational

MECHANISM

Activates the KISS1R (GPR54) receptor on GnRH neurons in the hypothalamus, triggering pulsatile GnRH release. Acts upstream of leuprolide-class drugs, which act directly at the pituitary.

INDICATIONS

- Investigational: hypogonadotropic hypogonadism
- Investigational: IVF oocyte-maturation trigger (alternative to hCG, lower OHSS risk)
- Investigational: low sexual desire — enhances limbic responses to sexual/romantic stimuli

DOSING · reported-use only, not recommended dosing; no approved indication

Diagnostic / IVF trigger	1–10 µg/kg IV/SC (GnRH test); 12.8 mg SC once (Phase 3 IVF trigger)
Off-label SC (libido/hormones)	1–4 nmol/kg (~0.1–0.3 mg), 1–2×/week — pulsatile may beat single dosing
Intranasal (experimental)	0.5–1 mg, 1–2 h before activity — direct CNS effect

EVIDENCE Phase 2/3 IVF triggering shows efficacy with reduced OHSS risk vs hCG. Imperial (Comninos/Dhilló) studies show kisspeptin enhances limbic-brain responses to sexual/romantic stimuli and raises LH and testosterone (peak ~30–120 min).

SAFETY Generally well-tolerated · Mild flushing and headache reported · Long-term safety not yet evaluated

ADVERSE Most common: mild flushing (most patients), transient headache, injection-site reactions. Generally well tolerated; no major safety signals in completed Phase 3 trials.

CLINICAL CAVEAT Acts UPSTREAM of GnRH (hypothalamus, not pituitary) — restoring physiologic pulsatility rather than overriding it like leuprolide. The near-term clinical lead is replacing hCG as an IVF trigger to reduce OHSS; libido/hormone-optimization use is experimental, with theoretical testosterone/PT-141 synergy.

KEY REFERENCES

- Comninos AN et al. JCI 2017;127:709
- Dhilló WS et al. JCEM 2005;90:6609 (PMID 16174713)
- Skorupskaitė K et al. (Anderson group) — multiple reproductive trials

Investigational (Phase 3, IVF trigger) · not approved

Oxytocin

Pitocin · Syntocinon

MECHANISM

Activates oxytocin receptors on uterine myometrium to drive contraction; also acts on mammary tissue (let-down reflex) and centrally on bonding-related neural circuits.

INDICATIONS

- Labor induction and augmentation
- Control of postpartum hemorrhage
- Off-label: autism social interaction trials (largely failed)

DOSING

IV labor induction	0.5–6 mU/min, titrate to contractions
IV/IM postpartum	10 units IM or 10–40 units IV diluted
Intranasal (off-label)	24 IU (up to 40) ~30–60 min before activity; ~4 IU/spray · <i>not FDA-validated</i>
Sublingual (off-label, weak evidence)	100–500 IU troche — slower onset, longer duration · <i>not FDA-validated</i>

EVIDENCE Cornerstone of labor management for >60 years. Autism, anxiety, and bonding trials have largely failed primary endpoints.

SAFETY Uterine hyperstimulation, fetal distress · Hypotension with rapid IV bolus · Water intoxication with prolonged high-dose use

ADVERSE Labor use: uterine hyperstimulation, fetal distress (most concerning), rare uterine rupture. Hypotension or arrhythmias with rapid IV bolus. Water intoxication with prolonged high-dose use. Nausea, vomiting; hypersensitivity rare.

CLINICAL CAVEAT Off-label CNS uses — autism, bonding, anxiety, sexual enhancement — have largely failed or are unvalidated; poor blood-brain-barrier penetration limits central effects of peripheral and intranasal oxytocin. Obstetric uses are the only evidence-based indications. For off-label sexual use, limit to 2–3×/week to avoid tolerance; compounded forms need refrigeration; synergy with PDE5 inhibitors or PT-141 is unproven.

KEY REFERENCES

- Carter CS et al. *Annu Rev Psychol* 2014;65:17
- Yatawara CJ et al. *Mol Psychiatry* 2016;21:1225

FDA-approved · EMA · all major regulators

GRADE A

VASOPRESSIN V2 RECEPTOR AGONIST

Desmopressin

DDAVP · *Stimate* · *Nocturna*

MECHANISM

Selective V2 receptor agonist on renal collecting duct cells; promotes aquaporin-2 insertion and water reabsorption. Also stimulates von Willebrand factor release from endothelial Weibel-Palade bodies.

INDICATIONS

- Central diabetes insipidus
- Nocturnal enuresis
- Mild hemophilia A and von Willebrand disease (boost vWF)
- Nocturia in adults

DOSING

Intranasal (DI)	5–40 µg/day in 1–2 divided doses
Oral (DI/enuresis)	0.1–0.4 mg/day
IV (vWD/hemophilia A)	0.3 µg/kg over 30 min

EVIDENCE Standard-of-care therapy for central DI for >40 years. Modest efficacy for nocturnal enuresis.

SAFETY Hyponatremia is the principal serious risk · Headache · Avoid in patients with type IIB vWD (paradoxical thrombocytopenia)

ADVERSE Most serious: hyponatremia (can be severe, with seizures). Headache, nausea, abdominal pain, facial flushing. Water intoxication if free water intake not restricted. Increased blood pressure (rare). Sodium monitoring required.

CLINICAL CAVEAT Sodium monitoring is essential — hyponatremia can be severe and is the principal serious AE. Type IIB von Willebrand disease is a contraindication (paradoxical thrombocytopenia).

KEY REFERENCES

- Robson WL et al. J Pediatr 1996;128:769
- Bertholini D, Butler GS. Br J Anaesth 2000

FDA-approved · all major regulators

THERAPEUTIC CLASS

Mitochondrial & Longevity

“Mitochondria-targeted peptides represent the frontier of longevity pharmacology – where the evidence is early, the mechanism is compelling, and the commercial stakes are enormous.”

8 ENTRIES

GRADE C

MITOCHONDRIAL-DERIVED PEPTIDE

MOTS-c

Research only

MECHANISM

Activates AMPK and modulates folate-methionine cycle; improves insulin sensitivity and mitochondrial function in preclinical models.

INDICATIONS

- None approved
- Investigational for metabolic dysfunction

DOSING · reported-use only, not recommended dosing; no approved indication

SC weekly total 5–15 mg/week, split into 2–3 doses (e.g., 5 mg M/W/F)

Microdosing (beginner) 3–5 mg SC 2×/week to assess tolerance

Cycling 4–8 weeks on / equal off; max 4 cycles/year

EVIDENCE Animal studies show improved exercise capacity, insulin sensitivity, and lifespan in mice. No human clinical trials published as of 2026.

SAFETY Limited human safety data · Anaphylaxis reported, especially with daily/high-frequency dosing · Theoretical concerns about modulating mitochondrial function long-term

ADVERSE Limited human data. Reported in early human studies: injection-site reactions, mild transient fatigue. No major safety signals to date. Long-term effects of modulating mitochondrial signaling pathways uncharacterized. Anaphylaxis reported with off-label use, particularly daily/supraphysiologic dosing.

CLINICAL CAVEAT Evidence base: preclinical mitochondrial biology + observational studies of endogenous MOTS-c; exogenous-administration human trials are only emerging in 2024–2025 — no Phase 3 data. Direct-to-consumer marketing for anti-aging and metabolic enhancement substantially outpaces the clinical data.

KEY REFERENCES

- Lee C, Cohen P et al. Cell Metab 2015;21:443 (PMID 25738461)
- Reynolds JC et al. Nat Commun 2021;12:470
- First human Phase 1 trials, 2024–2025

Research only · gray-market · July 2026 PCAC review (obesity, osteoporosis)

GRADE D

MITOCHONDRIAL-DERIVED NEUROPROTECTIVE PEPTIDE

Humanin

HN · research compound

MECHANISM

Anti-apoptotic: neutralizes pro-apoptotic Bcl-2 proteins (Bax, tBID). Activates pro-survival signaling via GPI30/WSX-1/CNTRF receptor complex (AKT, ERK1/2, STAT3). Extends *C. elegans* lifespan via autophagy.

INDICATIONS

- None approved
- Investigational: Alzheimer's, T2D, ischemic injury; gray-market for neuroprotection/anti-aging

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label SC (titration)

Wk 1–2: 0.5 mg 3×/week; week 3–4: 1 mg 4–5×/week; week 5+: 1.5–2 mg 5–7×/week

Potent variant

S14G-HNG used at much lower doses (~1000× potency)

EVIDENCE Extensive preclinical literature — anti-apoptotic in neurons (Hashimoto 2001), beta-cell protection (Muzumdar 2009), cardioprotection, lifespan extension in *C. elegans*. Circulating humanin levels correlate with healthspan. No human RCTs of exogenous administration published.

SAFETY No human safety data for exogenous administration · Anti-apoptotic mechanism raises theoretical oncologic concern with chronic dosing · Long-term effects unknown

ADVERSE Not characterized in humans. Animal studies have not shown major signals. Theoretical concern: pan-anti-apoptotic action could in principle support malignant-cell growth — no in-vivo evidence.

CLINICAL CAVEAT No RCTs of exogenous administration and no human efficacy evidence — preclinical anti-apoptotic and lifespan data plus observational human-level associations only. S14G-Humanin (HNG) is the studied variant; gray-market "humanin" may be either form. Off-label protocols: 8–12-wk cycles (12 on/4 off), AM dosing, effects in ~2–4 wks, sometimes stacked with NAD⁺ or metformin.

KEY REFERENCES

- Hashimoto Y et al. PNAS 2001;98:6336 (PMID 11371600)
- Hashimoto Y et al. J Neurosci 2001;21:9235 (PMID 11717354)
- Muzumdar RH et al. PLoS One 2009;4:e6334

Research only · gray-market · unapproved

SS-31 (Elamipretide)

Forzinity (Stealth BioTherapeutics)

MECHANISM

Accumulates in mitochondrial inner membrane via cardiolipin binding; preserves mitochondrial cristae structure, reduces ROS production, and maintains ATP synthesis under stress.

INDICATIONS

- Barth syndrome (FDA accelerated approval, ≥ 30 kg) — to improve muscle strength
- Investigational: primary mitochondrial myopathy, geographic atrophy, heart failure

DOSING

Barth (FDA-approved)	40 mg SC daily (≥ 30 kg)
Renal (eGFR < 30)	20 mg SC daily (not on dialysis)
Off-label mito (gray-market)	SC 2–5 mg 3–5 $\times$ /week, or intranasal 1 mg/day → 1–2 mg 2 $\times$ /day · <i>not FDA-validated</i>

EVIDENCE TAZPOWER (N=12, randomized crossover + 192-week open-label extension): primary endpoints (6-minute walk, fatigue score) NOT met. Secondary knee extensor muscle strength improved $>45\%$ in OL extension — basis for accelerated approval. Earlier MMPOWER-3 (primary mitochondrial myopathy) and ReNEW (geographic atrophy) missed primary endpoints.

SAFETY Injection-site reactions universal (100% in trial) · Hypersensitivity — skin and respiratory · Benzyl alcohol vehicle — NOT for neonates

ADVERSE Injection-site reactions universal (100%): erythema, pain, induration, pruritus, bruising, urticaria. Hypersensitivity — skin (rash) and respiratory (cough) — can occur weeks-to-months after start; manage with topical steroids and antihistamines.

CLINICAL CAVEAT First FDA-approved mitochondria-targeted therapeutic. Approval rested on an intermediate endpoint (knee-extensor strength) in a 12-patient trial where both primary endpoints (6MWT, fatigue) were not met; confirmatory trial required. Expansion beyond Barth is uncertain. Off-label mito-optimization (unvalidated): 8–12-wk cycles + 4-wk break, with meals, $\pm$ CoQ10/PQQ.

KEY REFERENCES

- Stealth BioTherapeutics. TAZPOWER
- Karaa A et al. MMPOWER-3. *Neurology* 2020;94:e29
- FDA label (NDA 215244, 2025)

FDA-approved (accelerated, Sept 19 2025) · Forzinity · Stealth BioTherapeutics

GRADE C

ANTI-AGING GLYCOPROTEIN (RECOMBINANT FRAGMENTS)

Klotho

Research compound · investigational

MECHANISM

α-Klotho is the obligate FGF23 co-receptor (phosphate/vitamin-D homeostasis), suppresses insulin/IGF-1 signaling (longevity), inhibits Wnt and TGF-β (anti-fibrotic), and activates Nrf2/FoxO antioxidant programs.

INDICATIONS

- None approved
- Investigational: CKD, neurodegeneration (ALS, AD), CV aging
- Off-label/gray-market: anti-aging, longevity

DOSING · reported-use only, not recommended dosing; no approved indication

SC (recombinant protein)	10–20 µg 2–3×/week — anti-aging (emerging, no validated protocol)
Intranasal (fragment)	50–100 µg daily — neuroprotection / cognitive
Oral (liposomal)	500–1000 µg daily

EVIDENCE Extensive preclinical data — renal, CV, and CNS benefits in animal models. No positive human efficacy data; Phase 1 anticipated (KLTO-202). Endogenous Klotho rises with exercise and certain drugs (ACE-I, statins, rapamycin).

SAFETY No human safety data for exogenous Klotho administration · Theoretical concern: altered phosphate handling and FGF23 signaling · Long-term effects unknown

ADVERSE Not characterized in humans. Animal studies have not shown major safety signals. Theoretical risks: hypophosphatemia (FGF23 axis modulation), altered calcium-vitamin D balance.

CLINICAL CAVEAT GRADE C reflects endogenous-Klotho biology — preclinical mechanism plus cohorts where Klotho tracks healthspan — NOT the marketed fragments, which have no human RCTs or proven bioequivalence. Off-label protocols run 3–6-month cycles (1-month breaks); intranasal is favored for cognitive use (~4–8 wks); monitor calcium, phosphate, and kidney function.

KEY REFERENCES

- Kuro-o M et al. Nature 1997;390:45 (PMID 9363890)
- Kurosuo H et al. Science 2005;309:1829 (PMID 16123266)
- Drew DA et al. JAMA 2017;317:846

Research only · gray-market · unapproved

GRADE D

KLOTHO-DERIVED PEPTIDE FRAGMENT

G-Klotho Frag 33

Research compound · marketed fragment

MECHANISM

Proposed: recapitulates a subset of Klotho actions — TGF- β receptor 2 binding (anti-fibrotic), restoration of endogenous Klotho via miR-223-3p/lncRNA-TUG1, Nrf2 antioxidant activation. Direct evidence for the 33-residue marketed form is absent.

INDICATIONS

- None approved
- Off-label/gray-market: anti-aging, renal protection claims, neuroprotection

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label SC 0.5–1 mg 2 $\times$ /week (vendor-suggested; no validated protocol)

Note Cycling commonly recommended by vendors despite no clinical basis

EVIDENCE No published trials of "G-Klotho Frag 33" specifically. Adjacent research on KP1 (Yuan 2022, Wei 2023) demonstrates anti-fibrotic and senescence-inhibiting effects in animal models of CKD. Marketing claims extrapolate from KP1 and full-length Klotho literature.

SAFETY No human safety data · Vendor sequence and purity unverified · Long-term effects unknown

ADVERSE Not characterized. Injection-site reactions, hypersensitivity, and contamination-related effects are the practical risks given uncertain manufacturing quality.

CLINICAL CAVEAT Evidence base: extrapolated from KP1 (Klotho-derived peptide 1) and full-length Klotho literature — the marketed "G-Klotho Frag 33" sequence is proprietary and not validated in peer-reviewed studies. Vendor sequences vary and bioequivalence to academic Klotho-derived peptides is not established. No published trials of this specific fragment.

KEY REFERENCES

- Yuan Q et al. Nat Commun 2022;13:438
- Wei J et al. (2023) Klotho-derived peptide 1 inhibits cellular senescence in fibrotic kidney via miR-223-3p/lncRNA-TUG1
- Doleckova I et al. Contipro

Research only · gray-market · vendor-defined product

GRADE C

TETRAPEPTIDE (PINEAL-DERIVED)

Epitalon

Research only

MECHANISM

Proposed to upregulate telomerase activity and extend telomere length in vitro. Mechanism in vivo poorly characterized; melatonin-modulating effects also claimed.

INDICATIONS

- None Western-approved
- Russia: gerontology/anti-aging applications
- Off-label/research: claimed effects on sleep quality and pineal-melatonin signaling (no Western RCT evidence)

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label SC (course)	5–10 mg daily × 10–20 days, repeated every 4–6 months; bedtime dosing — course-based, NOT continuous
Off-label oral	50–100 mg daily for 10–20 day courses (Khavinson capsule protocol)

EVIDENCE Studies almost entirely from Khavinson group; not independently replicated. Telomere-extension claims rest on in vitro work.

SAFETY Limited Western safety data · Reportedly well-tolerated in Russian use · Long-term effects unknown

ADVERSE Reported in Russian clinical and off-label use: generally well tolerated, occasional mild fatigue, sleep changes, transient injection-site reactions. Long-term safety in Western populations uncharacterized.

CLINICAL CAVEAT Evidence rests almost entirely on one Russian group (Khavinson et al.) — preclinical and small clinical studies only, with no independent Western replication or RCTs. Telomere-extension claims are based on in-vitro work that has not been independently confirmed.

KEY REFERENCES

- Anisimov VN et al. Biogerontology 2003;4:193 (PMID 14732212)
- Khavinson VKh, Morozov VG. multiple Russian publications, 1980s–2010s

Research only outside Russia · July 2026 PCAC review (insomnia)

GRADE D

NNMT INHIBITOR (SMALL MOLECULE)

5-Amino-1MQ

Research only · gray-market

MECHANISM

Inhibits NNMT, the enzyme that methylates nicotinamide. NNMT inhibition raises intracellular NAD⁺ and lowers fat-storage gene expression in adipocytes; rodent studies show reduced adiposity and improved metabolic markers.

INDICATIONS

- None approved
- Off-label/gray-market: weight loss, "metabolic health" enhancement

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label oral 50–150 mg daily

Typical cycle 30-day on / 30-day off

EVIDENCE Strong rodent data (Sinclair lab and others) for reduced adiposity and improved metabolic markers in obese mice. No published human clinical trials.

SAFETY No human safety data · NNMT involved in stem-cell regulation — chronic inhibition effects unknown · Theoretical effects on methyl-donor balance

ADVERSE No published human safety data. Anecdotal reports of GI upset and mild headache. Long-term effects of NNMT inhibition uncharacterized in humans.

CLINICAL CAVEAT A small molecule, not a peptide — included here because it is marketed and sold through the same gray-market channels as injectable peptides. NNMT inhibition has compelling preclinical metabolic data but zero human clinical trials. Trial stage: preclinical only (rodent metabolic studies).

KEY REFERENCES

- Neelakantan H et al. *Biochem Pharmacol* 2018
- Kraus D et al. (Cleveland Clinic) preclinical metabolic studies

Research only · gray-market (small molecule, not approved)

GRADE D

D-RETRO-INVERSO SENOLYTIC PEPTIDE

FOXO4-DRI

Proxofim · FOXO4-DRI · research compound (senolytic)

MECHANISM

In senescent cells, FOXO4 sequesters p53 in the nucleus, preventing apoptosis. The DRI peptide disrupts this interaction selectively in senescent cells (which express high FOXO4), releasing p53 to the mitochondria to trigger intrinsic apoptosis. Normal cells are spared — more selective than dasatinib + quercetin.

INDICATIONS

- None approved
- Investigational: senolytic therapy for aging-related dysfunction, frailty, sarcopenia, fibrotic disease
- Off-label/gray-market: longevity-clinic use without human evidence

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label SC (initial cycle) 2–5 mg daily × 3–5 days; then 1–3 weeks off

Advanced cycle 5–10 mg daily × 5–7 days; 1 week on, 1–3 weeks off (extrapolated from ~5 mg/kg mouse dosing)

EVIDENCE One landmark mouse study (Baar et al., Cell 2017): >50% senescent-cell clearance with restored fur density, improved renal function, and testosterone returning toward youthful levels. No human trials. Senolytic translation has been poor — dasatinib + quercetin showed only modest human Phase 1 effects.

SAFETY No human safety data · Theoretical risks if normal cells are inadvertently targeted · Cost-prohibitive for routine use (\$1000s per cycle in gray-market)

ADVERSE Not characterized in humans. Mouse studies showed transient fur loss during dosing cycles. Theoretical risk of immune-cell or stem-cell depletion has not been observed in animals.

CLINICAL CAVEAT No published human trials; preclinical mouse data is striking but senolytic translation has been disappointing. Intermittent cycling (1 wk on, 1–3 off) is considered essential to allow immune clearance of apoptotic cells; no validated biomarker tracks senescent-cell clearance in humans. Gray-market use at \$1000s/cycle without human safety or efficacy data carries unjustifiable risk. Preclinical only.

KEY REFERENCES

- Baar MP et al. Cell 2017;169:132 (PMID 28340339)

Research only · no human trials · not approved anywhere

THERAPEUTIC CLASS

Neurological Peptides

“The blood-brain barrier is the defining pharmacological challenge of neuropeptide medicine – and the primary reason intranasal delivery routes have attracted sustained research attention.”

7 ENTRIES

GRADE C

ANXIOLYTIC PEPTIDE

Selank

Approved in Russia

MECHANISM

Synthetic heptapeptide (TP-7): the tuftsin tetrapeptide (Thr-Lys-Pro-Arg) plus a Pro-Gly-Pro extension for stability and CNS activity. Modulates GABAergic and serotonergic systems, raises BDNF, and retains tuftsin-like immunomodulation (IL-6, immunoglobulin expression).

INDICATIONS

- Anxiety (Russia)
- Off-label: cognitive enhancement (attention, memory, learning), immune modulation

DOSING

Intranasal (Russia-approved) 0.15% solution, 250–500 µg 2–3×/day

Off-label SC 250–500 µg daily · *not FDA-validated*

EVIDENCE Russian RCTs (non-Western journals) report anxiolysis comparable to benzodiazepines without sedation, impairment, or dependence, plus nootropic gains in attention and memory; few independent replications. Animal data is more consistent.

SAFETY Generally well-tolerated in Russian use · Long-term Western safety data lacking · Limited drug-interaction data

ADVERSE Generally well tolerated in Russian clinical use. Reported: occasional drowsiness, dry mouth, nasal irritation (intranasal route). No serious AEs documented; long-term Western data lacking.

CLINICAL CAVEAT Approved in Russia (2009) as an intranasal anxiolytic/nootropic; Russian studies report benzodiazepine-comparable anxiety relief without sedation, cognitive impairment, or dependence — but no Western RCTs exist. Intranasal gives rapid CNS access (effects in ~15–30 min); despite a minutes-long half-life, effects persist for hours via downstream signaling. Used daily in 14–30-day courses without tolerance or withdrawal. Western efficacy and interaction data are lacking.

KEY REFERENCES

- Russian publications (Inozemtsev, Semenova) — anxiolytic preclinical and small clinical studies

Approved in Russia · research elsewhere

GRADE C

ACTH(4-10) ANALOGUE

Semax

Approved in Russia

MECHANISM

Synthetic ACTH(4-10) fragment plus a Pro-Gly-Pro extension for metabolic stability; lacks corticotropic activity (no cortisol rise). Modulates BDNF, NGF, and TrkB signaling — enhancing neuroplasticity, memory, and neuroprotection (raises BDNF 3–8× in animals, among the highest of any agent).

INDICATIONS

- Stroke recovery, cognitive disorders, encephalopathy, optic-nerve disease, childhood ADHD (Russia)
- Off-label cognitive enhancement

DOSING

Intranasal 0.1% (cognitive)	200–600 µg 2–3×/day
Intranasal 1% (stroke/neuroprotection)	1–3 mg 2–3×/day
Off-label SC	200–500 µg daily · <i>not FDA-validated</i>

EVIDENCE Russian trials report benefit in stroke recovery (given within ~6 h), cognitive enhancement, optic-nerve disease, and childhood ADHD; quality and independent replication are limited. Animal data is more consistent.

SAFETY Generally well-tolerated in Russian clinical use · Long-term Western data lacking

ADVERSE Generally well tolerated. Reported: nasal irritation (intranasal route), occasional mild anxiety or restlessness, rare reports of insomnia. Long-term safety in Western populations not characterized.

CLINICAL CAVEAT Approved in Russia (2011) as a nasal spray for cognition, stroke recovery, and neuroprotection; ACTH(4-10)-derived but without corticotropic activity. Rapid CNS access (effects ~15–30 min); used in 10–14-day courses with breaks. A more potent acetylated variant, N-Acetyl Semax (NASA), has greater BBB penetration — use lower doses. No Western RCTs; efficacy rests on non-Western evidence.

KEY REFERENCES

- Russian publications (Myasoedov group) — stroke, cognitive enhancement

Approved in Russia · research elsewhere · July 2026 PCAC review (cerebral ischemia, migraine, trigeminal neuralgia)

GRADE D

ANGIOTENSIN IV ANALOGUE (NEUROACTIVE)

Dihexa

Research only · gray-market

MECHANISM

Hepatocyte growth factor (HGF) / c-Met signaling activator with reported synaptogenic effects in preclinical models (dendritic arborization, neural connectivity). Reported ~10 million times more potent than BDNF in some neurite-outgrowth assays.

INDICATIONS

- None approved
- Off-label/gray-market: nootropic, cognitive enhancement; investigational for neurodegenerative disease

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label oral (AM)	Conservative 1–5 mg; standard 5–10 mg; advanced 10–20 mg daily
Transdermal cream	20 mg/g, applied daily (AM)
Intranasal spray (compounded)	Formulation-dependent, 1–2×/day

EVIDENCE Rodent cognitive benefit in MCI/Alzheimer models (Wright/Harding labs) — but only in impaired animals (≈ 1.44 – 2.88 mg/kg), not normal baseline. The dihexa prodrug fosgonimeton (ATH-1017) FAILED Phase 2/3 in Alzheimer despite Phase 1 biomarker signals. No human trials of dihexa itself.

SAFETY No human safety data for dihexa · c-Met is an oncogene — chronic activation raises unaddressed tumor/metastasis concerns · Very long half-life (~12 days) → prolonged exposure of unknown consequence

ADVERSE Limited human data. Reported by users: mild headache, transient fatigue, occasional anxiety or insomnia at higher doses. No serious AEs documented in animal studies; oncogenic risk untested.

CLINICAL CAVEAT Once promising on rodent data, but the dihexa prodrug fosgonimeton failed Phase 2/3 in Alzheimer; there are no human efficacy studies of dihexa itself. c-Met is an oncogene — chronic activation raises unaddressed tumor/metastasis concerns, and the ~12-day half-life means prolonged exposure with unknown consequences. No validated human protocol — use is uncontrolled self-experimentation.

KEY REFERENCES

- McCoy AT et al. J Pharmacol Exp Ther 2013;344:141
- Athira Pharma. LIFT-AD

Research only · gray-market

GRADE D

PINEAL-DERIVED TRIPEPTIDE

Pinealon

Research compound (Russia)

MECHANISM

Khavinson bioregulator (tripeptide): proposed DNA-level gene regulation supporting pineal function, melatonin production, circadian rhythm, and neuroprotection. Mechanism not characterized in mainstream literature.

INDICATIONS

- None Western-approved
- Russian/off-label: sleep disorders, age-related pineal decline, cognitive support, neuroprotection, circadian and neuroendocrine health

DOSING · reported-use only, not recommended dosing; no approved indication

Oral standard 20 mg daily, evening — brain health and sleep

Oral intensive 20 mg 2×/day for 2 months, then daily — cognitive decline / sleep disorders

EVIDENCE All published studies originate from the Khavinson group; not independently replicated outside Russia. In vitro evidence for antioxidant effects.

SAFETY Limited Western safety data · Reportedly well-tolerated in Russian use · Long-term safety uncharacterized

ADVERSE Generally well tolerated in Russian use. Reported: mild headache, occasional sleep changes. No serious AEs documented; long-term Western data lacking.

CLINICAL CAVEAT Evidence rests entirely on the Khavinson cytomedine group — no Western trials, no human RCTs, and no independent replication. Oral courses run 2–3 months (effects in ~2–4 wks, reportedly persisting 4–6 months), repeated every 4–6 months; evening dosing supports circadian rhythm, and it is often paired with Cortagen. One of several single-group bioregulators (with Epitalon) for which independent confirmation remains lacking.

KEY REFERENCES

- Khavinson VKh group — Russian publications only

Research only outside Russia

GRADE B

PORCINE BRAIN PEPTIDE HYDROLYSATE

Cerebrolysin

Cerebrolysin (Ever Pharma)

MECHANISM

Proposed to mimic the action of endogenous neurotrophic factors (BDNF, NGF, GDNF). Demonstrated effects on neurite outgrowth, neurogenesis, and neuronal survival in preclinical models.

INDICATIONS

- Stroke recovery (in approved countries)
- Vascular dementia
- Alzheimer disease (adjunct)
- Traumatic brain injury
- Pediatric developmental disorders

DOSING

IV	10–30 mL daily/every other day in 10–20 day courses; 20–50 mL daily for 7–14 days in acute stroke or TBI
IM	1 mL IM daily for 40–80 days (outpatient alternative)

EVIDENCE Alvarez 2011 Phase 3 (n=133, Alzheimer): CIBIC+ improved across 10/30/60 mL doses at 24 wks. Stroke and TBI RCTs (CASTA, CARS-1/2) show modest functional improvement; 2023 Cochrane: likely no benefit for all-cause mortality in acute ischemic stroke.

SAFETY Generally well-tolerated · Hypersensitivity, vertigo, headache reported · Contraindicated in severe renal impairment and epilepsy

ADVERSE Most common: headache, vertigo, agitation, sweating, hyperhidrosis. Hypersensitivity reactions (rare). Injection-site irritation at IV catheter. Generally well tolerated in long-term use.

CLINICAL CAVEAT One of the very few peptide preparations with positive (if heterogeneous) RCT evidence in stroke recovery. The heterogeneous-mixture nature rather than a defined peptide complicates pharmacology — the exact active component remains unidentified.

KEY REFERENCES

- Heiss WD et al. Stroke 2012;43:630 (PMID 22282884)
- Bornstein NM et al. Acta Neurol Scand 2018

Approved in 50+ countries (notably China, Russia, Mexico, Austria) · not FDA-approved

GRADE D

SLEEP-MODULATING NEUROPEPTIDE

DSIP (Delta Sleep-Inducing Peptide)

Research only · gray-market

MECHANISM

Sleep-modulating nonapeptide; modulates sleep-wake cycles via GABAergic, glutamatergic, and opioid systems and influences cortisol rhythms and stress response. No specific cognate receptor identified despite decades of study; very short half-life (~7–8 min, rapidly degraded by aminopeptidases), requiring IV/SC dosing.

INDICATIONS

- None approved
- Off-label gray-market: insomnia, withdrawal syndromes (alcohol, opioid), chronic pain, narcolepsy

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label SC at bedtime 100–500 µg

Typical pattern Intermittent dosing, not nightly

EVIDENCE Small 1980s trials suggested improved subjective sleep quality, reduced sleep latency, and increased delta-wave EEG activity in chronic insomnia; also explored for narcolepsy, alcohol withdrawal, and analgesia. No modern controlled RCTs.

SAFETY Generally well-tolerated in limited use · Long-term safety not characterized

ADVERSE Limited human data. Reported: vivid dreams, mild morning grogginess at higher doses, injection-site reactions. No serious AEs documented.

CLINICAL CAVEAT One of the original 'designer sleep peptides' with intriguing 1970s–80s data that never translated into approved therapy. Modern sleep pharmacology (orexin antagonists) has more rigorously studied alternatives. Trial stage: small 1970s–80s clinical trials; never reached registration; no modern trials.

KEY REFERENCES

- Schoenenberger GA, Monnier M. Proc Natl Acad Sci USA 1977;74:1282
- No modern clinical trials available

Research only · gray-market · July 2026 PCAC review (opioid withdrawal, chronic insomnia, narcolepsy)

GRADE D

TREK-1 CHANNEL INHIBITOR PEPTIDE

PE-22-28

Research compound (spadin analog)

MECHANISM

Selectively inhibits TREK-1 (TWIK-related potassium channel 1), a background K⁺ channel expressed in serotonergic and cortical neurons. TREK-1 inhibition is hypothesized to produce antidepressant effects with faster onset than SSRIs by disinhibiting serotonergic signaling.

INDICATIONS

- None approved
- Investigational: depression, treatment-resistant depression
- Off-label intranasal use in some longevity/wellness clinics

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label intranasal 400 µg per spray, 2 sprays per nostril daily (≈1.6 mg/day total)

EVIDENCE Preclinical only — rodent models support antidepressant activity via TREK-1 inhibition. Mendelian-validation evidence: TREK-1 polymorphisms are associated with treatment-resistant depression in humans. No published human clinical trials of PE-22-28 itself.

SAFETY No human safety data · Theoretical cardiac concern (TREK-1 expressed in atrial myocytes) · Long-term effects unknown

ADVERSE Not characterized in humans. Rodent studies have not shown significant adverse effects but the trial corpus is small. Theoretical risk of arrhythmia given cardiac TREK-1 expression.

CLINICAL CAVEAT Mechanistically interesting — TREK-1 is a validated antidepressant target — but PE-22-28 itself has no human clinical evidence. The intranasal route is convenient but unstudied; cardiac TREK-1 expression is a theoretical safety concern for chronic dosing. Trial stage: preclinical only.

KEY REFERENCES

- Djillani A et al. Front Pharmacol 2019;10:643

Research only · no human trials

THERAPEUTIC CLASS

Investigational Oncology

“Cancer-selective membrane disruption and opioid receptor modulation represent two distinct peptide strategies in early oncological investigation, both currently lacking pivotal human trial data.”

2 ENTRIES

PNC-27

Research only · gray-market

MECHANISM

Targets HDM-2 (MDM2) expressed on cancer-cell plasma membranes. The peptide forms pores in tumor-cell membranes (cell-selective necrosis/lysis), sparing normal cells which lack surface HDM-2. Mechanism is distinct from p53-stabilizing MDM2 inhibitors.

INDICATIONS

- None approved
- Investigational/off-label: various cancers (forum and clinic use)

DOSING · reported-use only, not recommended dosing; no approved indication

Off-label SC 200–300 µg SC three times weekly (gray-market protocol)

EVIDENCE Preclinical only — multiple cell-line and animal studies. No published Phase 1 human safety data. Clinical efficacy claims rest entirely on in vitro and rodent work.

SAFETY No human safety profile characterized · Off-label injectable use carries undefined risks · Theoretical immunogenicity from 32-residue synthetic peptide

ADVERSE No characterized human safety data. Gray-market reports describe injection-site reactions, fatigue, fevers. Tumor-lysis-style syndromes are theoretically possible but not documented.

CLINICAL CAVEAT Often marketed online as a 'cancer cure' — there is no human clinical evidence to support this claim. Compelling mechanism and preclinical data have not yet translated to human trials. Use outside an IRB-approved protocol is not supported by the evidence. Trial stage: preclinical only; no human trials.

KEY REFERENCES

- Michl J et al. Cancer Chemother Pharmacol 2006;58:355
- No human trials published

Research only · not approved anywhere · no human trial data

GRADE D

ENDOGENOUS OPIOID PENTAPEPTIDE

Met-5-Enkephalin

Opioid Growth Factor · OGF

MECHANISM

Activates OGF_R (opioid growth factor receptor, nuclear-shuttling), inducing p16 and p21 cyclin-dependent kinase inhibitors. Inhibits cell proliferation in OGF_R-expressing tumors and immune cells. Mechanistically distinct from classical opioid analgesia.

INDICATIONS

- None approved
- Investigational: pancreatic cancer (Phase 2, modest activity)
- Off-label/research: multiple sclerosis, Crohn disease (same axis as low-dose naltrexone)

DOSING · reported-use only, not recommended dosing; no approved indication

Investigational IV 10 mg/m² IV weekly (Zagon Phase 2 pancreatic cancer)

Off-label SC 10 mg SC 2×/day, 2×/week

EVIDENCE Phase 2 in pancreatic cancer (Smith et al., Open Access Journal of Clinical Trials 2010) showed median survival ~8 mo — disappointing relative to chemotherapy. The OGF_R-mediated mechanism is real and reproducible. Low-dose naltrexone (LDN) achieves the same receptor effect with better pharmacokinetics.

SAFETY Generally well tolerated in trials · IV pulse causes mild flushing and hypotension · Long-term safety not characterized

ADVERSE Flushing, fatigue, mild GI upset reported in trials. Theoretical concern about delayed wound healing given anti-proliferative effects on normal cells.

CLINICAL CAVEAT The OGF_R mechanism is well-characterized, but met-enkephalin's plasma half-life of minutes makes direct dosing inefficient. Low-dose naltrexone (LDN, 1.5–4.5 mg PO nightly) transiently blocks OGF_R, producing rebound pathway activation with oral convenience — and has displaced direct enkephalin dosing in most protocols. Phase 2 in pancreatic cancer (Smith 2010) showed modest activity; not advanced.

KEY REFERENCES

- Smith JP et al. Open Access J Clin Trials 2010;2:37

Research only · not approved as cancer therapy

THERAPEUTIC CLASS

GI & Other

“Linaclotide's success in IBS-C validated a new therapeutic principle: a peptide that acts entirely within the gut lumen, achieving clinical benefit without systemic absorption.”

3 ENTRIES

GRADE A

GUANYLATE CYCLASE-C AGONIST

Linaclotide

Linzess · Constella

MECHANISM

Activates intestinal guanylate cyclase-C (GC-C) on the luminal surface of enterocytes; raises intracellular cGMP, opens CFTR, drives chloride and water secretion into the intestinal lumen. Local action; minimal systemic absorption.

INDICATIONS

- IBS with constipation (IBS-C)
- Chronic idiopathic constipation (CIC)
- Functional constipation in pediatric patients (ages 6+)

DOSING

Oral daily (CIC)	145 µg daily
Oral daily (IBS-C)	290 µg daily
Pediatric FC (6–17 yrs)	72 µg daily

EVIDENCE Multiple pivotal Phase 3 trials show significant improvement in stool frequency, abdominal pain, and global IBS-C symptom scores.

SAFETY Diarrhea is the main dose-limiting effect · Contraindicated in pediatric patients < 2 years (lethal dehydration in juvenile mice) · GI obstruction must be ruled out

ADVERSE Most common: diarrhea (16–20%, dose-limiting), abdominal pain (7%), flatulence, abdominal distension, viral gastroenteritis, headache. Severe diarrhea may require dose interruption. Boxed warning for fatal dehydration in pediatric patients < 2 years (juvenile-mouse data).

CLINICAL CAVEAT Take 30 minutes before the first meal of the day for best effect — food substantially reduces absorption. Diarrhea is dose-limiting; consider starting with the lower 145 µg dose even for IBS-C in sensitive patients.

KEY REFERENCES

- Lembo AJ et al. NEJM 2011;365:527 (PMID 21830967)
- Chey WD et al. Am J Gastroenterol 2012;107:1702 (PMID 22986440)

FDA-approved (Linzess US 2012, Constella EU) · EMA

LL-37 (Cathelicidin)

Research only

MECHANISM

Direct antimicrobial via membrane disruption (broad-spectrum). Also modulates innate immune cell function, wound healing, and angiogenesis through formyl peptide receptor 2 and other targets.

INDICATIONS

- None approved
- Investigational: chronic wounds, infections, autoimmune skin disease (where its dysregulation is pathogenic)

DOSING · reported-use only, not recommended dosing; no approved indication

Investigational topical

No standardized human protocol

EVIDENCE Substantial preclinical data; some early-phase human studies in chronic wounds. Dual role (host defense vs autoimmune driver in rosacea, psoriasis) complicates therapeutic development.

SAFETY Limited human safety data · Pro-inflammatory role in some skin diseases · Cytotoxicity at higher concentrations

ADVERSE Limited human data. Topical or trial use: skin irritation, erythema. Pro-inflammatory role implicated in rosacea, psoriasis, and atopic dermatitis pathogenesis — meaning therapeutic use must navigate a dual role. Cytotoxicity at high concentrations.

CLINICAL CAVEAT Evidence base: extensive preclinical antimicrobial and oncology literature + early Phase 1 melanoma data — no positive Phase 2 or III published. Dual role complicates therapy — antimicrobial and wound-healing on one hand, pro-inflammatory in rosacea, psoriasis, and atopic dermatitis on the other. No approved LL-37 therapeutic exists. Trial stage: preclinical and early Phase 1 oncology; no Phase 2 completed.

KEY REFERENCES

- Zanetti M et al. FEBS Lett 1995;374:1
- Multiple preclinical oncology papers; early Phase 1 melanoma data

Research only · investigational topical · no human therapeutic approval

GRADE C

TIGHT JUNCTION REGULATOR

Larazotide acetate

Investigational (9 Meters Biopharma)

MECHANISM

Antagonizes zonulin, the endogenous regulator of intestinal tight junctions; reduces gluten-induced permeability and downstream inflammation.

INDICATIONS

- Investigational for celiac disease
- Off-label gray-market use for "leaky gut" complaints

DOSING · *reported-use only, not recommended dosing; no approved indication*

Trial protocols 0.5–1 mg orally 3×/day before meals

EVIDENCE Earlier Phase 2b trials showed reduced symptoms vs placebo, but pivotal Phase 3 CeDLara failed.

SAFETY Headache common · Generally well-tolerated · Long-term safety not established

ADVERSE Most common in trials: headache (12%), upper respiratory tract infection (10%), urinary tract infection, fatigue. Generally well tolerated; failed Phase 3 on efficacy rather than safety grounds.

CLINICAL CAVEAT Evidence base: pivotal Phase 3 CeDLara failed its primary endpoint (2022); off-label "leaky gut" use rests on mechanistic plausibility and anecdote, not RCT evidence. The pivotal Phase 3 CeDLara trial (2022) failed its primary endpoint. Off-label "leaky gut" marketing substantially exceeds the actual evidence base.

KEY REFERENCES

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Investigational · not approved

Clinical pearls

A THREE-TIER PRESCRIBING FRAMEWORK

Tier 1

FDA-approved, on-label. Highest evidence; default when an approved agent fits the need. *Semaglutide for T2DM · tirzepatide for obesity · tesamorelin for HIV-lipodystrophy.*

Tier 2

FDA-approved, off-label. Legal; document rationale, obtain informed consent. *Tesamorelin for non-HIV visceral adiposity · sermorelin for adult use outside approved pediatric GHD.*

Tier 3

Non-approved · compounded or research. Elevated risk; source 503A/503B with COA, monitor closely, lower threshold to discontinue. *BPC-157 · TB-500 · CJC-1295 · ipamorelin · GHK-Cu.*

FOUNDATIONS BEFORE PEPTIDES

Optimize the substrate first. Peptides amplify a healthy foundation — diet, sleep, hydration, regular exercise — they do not substitute for one. **Start low, go slow.** Begin at the lowest effective dose and titrate over weeks. **Don't stack too many at once.** Each new agent confounds attribution of effect and AEs — isolate variables. **Monitor for anaphylaxis** across all peptide classes, particularly with daily or supraphysiologic dosing; counsel patients on signs and document an emergency response plan.

ELEVEN PEARLS FOR THE WORKING CLINICIAN

PEARL 1

Start low, go slow with GLP-1s. Semaglutide 0.25 mg/wk × 4 wks, titrate q4wk; tirzepatide 2.5 mg/wk start. Smaller meals, lower-fat foods, hydration, ondansetron PRN. Discontinuation at starting dose under-tests the drug.

PEARL 2

IGF-1 is your biomarker for GH secretagogues. Target upper-half age-adjusted range; do not exceed ULN. Check baseline → 4–6 wks → q3mo. Reduce dose or lengthen interval before discontinuing.

PEARL 3

Source matters — compounded is not all the same. Insist on 503A/503B with COA: identity, purity ≥ 98% by HPLC, sterility, endotoxin. One online-vendor case showed 62% actual peptide content with bacterial contamination causing injection-site abscess.

PEARL 4

Preserve lean mass during GLP-1 weight loss. STEP 1 found ~40% of weight lost was lean body mass. Prescribe progressive resistance training and protein 1.2–1.6 g/kg/day. Concurrent GH secretagogue use is unvalidated and may worsen glycemic control.

PEARL 5

Cycle wellness peptides. Typical: 8–12 weeks on, 4–8 weeks off. Reassess clinical endpoints at each cycle's end rather than perpetuating open-ended prescriptions.

PEARL 6

Know the FDA 503A Category 2 bulk substances list (Sept 2023). Seventeen peptides restricted from 503A compounding — BPC-157, TB-500, CJC-1295, ipamorelin, AOD-9604, GHK-Cu, melanotan II, thymosin α 1, KPV, others. Legal landscape evolving — verify before prescribing.

PEARL 7

Recognize red flags and know when to stop. GLP-1: severe abdominal pain (pancreatitis), neck mass (MTC workup), hypersensitivity. GH secretagogues: IGF-1 above ULN, acromegaloïd features. *Compounded:* fever, injection-site infection, LFT/RFT derangement.

PEARL 8

Document informed consent in detail for off-label and gray-market use. Off-label use of FDA-approved drugs is legal; gray-market peptides carry elevated risk. Document the GRADE of evidence, regulatory status, alternatives offered, expected effects, monitoring plan, and discontinuation criteria. Verbal-only consent is professionally vulnerable; written documentation protects clinician and patient.

PEARL 9

Treat pregnancy and lactation as contraindications by default. Most peptides have no FDA pregnancy category; almost none have lactation data. Discontinue at conception planning for non-essential peptides. Exceptions are specifically labeled (e.g., insulin is mandatory in pregnancy; teriparatide is contraindicated). Counsel reproductive-age patients before initiation.

PEARL 10

Set the baseline before starting — and track outcomes. Document baseline for each peptide's intended effect: weight/waist for GLP-1 RAs, IGF-1 for GH secretagogues, validated libido scales for melanocortins, energy/sleep symptom scores for mitochondrial peptides. Re-check at 8–12 weeks. If no measurable change against baseline, stop the drug rather than escalating dose.

PEARL 11

Dose GH secretagogues at night, on an empty stomach. Best taken at night, 2–3 h after the last meal, on an empty stomach — this synergizes with the natural GH surge during sleep. Counsel patients that body-composition effects appear gradually, over 3–6 months.

Neuropsychiatric & addiction signals in GLP-1 therapy

A cross-class summary, collected once rather than repeated in each entry. All signals below are hypothesis-generating; none is an approved indication.

WHY A REWARD-PATHWAY SIGNAL IS PLAUSIBLE

GLP-1 receptors are expressed in mesolimbic reward circuitry — the ventral tegmental area and nucleus accumbens — where activation modulates dopaminergic signaling. This gives a mechanistic rationale for the recurring observation that GLP-1 receptor agonists may reduce alcohol and nicotine consumption and blunt some reward-driven behaviors. Plausibility is not efficacy: the human evidence remains early.

ALCOHOL USE DISORDER (AUD)

The strongest prospective signal is for **semaglutide**, now in dedicated trials (e.g., SEMALCO) after observational and early clinical data suggested reduced intake. **Tirzepatide** shows an AUD-reduction signal exceeding other agents in observational datasets, but heavily confounded. **Dulaglutide** (Probst et al., JCI Insight 2023) was linked to reduced alcohol use during a smoking-cessation study. **Liraglutide** and **exenatide** show preclinical/observational signals.

SMOKING CESSATION

The clearest randomized signal is for **exenatide**: one positive RCT in combination with a nicotine patch. Other agents have observational or preclinical data only.

INTERPRETATION FOR PRESCRIBERS

None of these is an approved indication, and none should drive drug selection today. The class-wide consistency is worth knowing — especially when a patient with obesity or T2DM also has AUD or tobacco dependence — but dedicated Phase 3 trials are still in progress. Treat any addiction benefit as a possible secondary observation, not a reason to prescribe.

Receptor desensitization & peptide cycling

Why many peptides lose effect with continuous use — and why scheduled off-cycles ("peptide holidays") restore it.

Many peptide effects fade with sustained or frequent dosing (tachyphylaxis), usually from adaptive down-regulation of the target receptor — sometimes compounded by counter-regulatory feedback or anti-drug antibodies. Matching dosing to the body's natural rhythm and scheduling off-cycles preserves the response.

HOW DESENSITIZATION HAPPENS

- **Receptor uncoupling & internalization.** Agonist binding recruits GPCR kinases and β -arrestin, which uncouple the receptor from its G-protein and draw it into the cell (endocytosis); sustained signaling then lowers receptor number (down-regulation).
- **Exposure pattern.** Pulsatile, physiologic signaling preserves sensitivity; continuous or supraphysiologic exposure (long-acting analogues, frequent high doses) desensitizes fastest — hence pulsatile GH secretagogues that mimic the overnight rhythm hold up better.
- **Counter-regulatory feedback.** The body opposes sustained stimulation — e.g., somatostatin tone rises against continuous GH-axis drive — blunting response independent of the receptor.
- **Anti-drug antibodies.** Repeated exposure to a foreign peptide can raise neutralizing antibodies that cut efficacy or, rarely, cause hypersensitivity — likelier with impure gray-market product.

CLASS EXAMPLES

- **GH secretagogues.** Hexarelin desensitizes fastest — GH responses can fade within 2–4 wks of daily use, requiring cycling ($\approx$ 4 wks on, 2–4 wks off); ipamorelin and CJC-1295 are slower but cycling is still advised.
- **GnRH agonists** (leuprolide, goserelin). Desensitization is the therapeutic goal — continuous dosing down-regulates pituitary GnRH receptors, suppressing LH/FSH after an initial flare. Melanocortins and others show variable tachyphylaxis, arguing for on-demand or spaced use.

WHY PEPTIDE HOLIDAYS

Scheduled off-cycles let receptor density and signaling recover, restoring response and limiting antibody formation. Typical: 8–12 wks on, 4 wks off — shorter for fast-desensitizing agents (hexarelin $\approx$ 4 on / 2–4 off). Treat a waning effect as the cue to cycle rather than escalating the dose.

Lab & monitoring reference

Baseline labs, on-treatment monitoring, drug interactions, and stopping criteria.

BASELINE PANEL

- Comprehensive metabolic panel · fasting glucose · HbA1c · fasting lipid panel
- IGF-1 (if any GH secretagogue or GHRH analogue)
- Thyroid function (TSH, free T4) · complete blood count · hs-CRP
- Fasting insulin (if metabolic-syndrome workup)
- Sex hormones (total + free testosterone, estradiol, DHEA-S) when contextually relevant
- **For GLP-1 candidates additionally:** personal & family history of MTC or MEN2 (contraindication); diabetic retinopathy screen if T2DM
- **DEXA (body composition)** at baseline for recomposition or anti-aging goals
- **Telomere length** at baseline (and at follow-up) for Epitalon users — telomerase claims are unproven, so interpret with caution

MONITORING DURING THERAPY

GLP-1 and incretin agents — every 3 months

- HbA1c, fasting glucose, lipid panel, hepatic and renal function
- Screen symptoms: pancreatitis, gallbladder disease, retinopathy progression (T2DM)

GH secretagogues and GHRH analogues

- IGF-1: baseline → 4–6 weeks → every 3 months. Target upper-half of age-adjusted reference range; do not exceed ULN
- Fasting glucose + HbA1c every 3 months (GH stimulates insulin resistance)

Regenerative peptides (BPC-157, TB-500, KPV, GHK-Cu, etc.)

- No validated monitoring biomarker. Periodic LFTs and RFTs
- Assess injection-site reactions, signs of immune activation, and clinical endpoints relevant to the treatment indication

DRUG-INTERACTION NOTES

- Low CYP450 potential (peptides are metabolized by proteolysis, not hepatic enzymes).
Exception — GLP-1 agonists slow gastric emptying: may alter absorption of narrow-window oral drugs (levothyroxine, OCs, warfarin) — time-stagger or monitor levels

STOPPING CRITERIA · RED FLAGS

- **GLP-1 / incretins:** persistent severe abdominal pain (pancreatitis) · neck mass, dysphagia, hoarseness (MTC) · severe hypersensitivity
- **GH secretagogues:** IGF-1 above age-adjusted ULN · new-onset diabetes · acromegaloïd features (edema, arthralgia, carpal-tunnel symptoms)
- **Compounded peptides:** fever, injection-site infection or abscess · new rash or systemic symptoms · LFT or RFT derangement

Anti-doping urine & plasma testing

What gets tested, where, and what cannot.

DIRECT VS INDIRECT DETECTION

Exogenous synthetic peptides (GHRPs, GHRH analogues, melanocortin agonists, BPC-157, TB-500) are detected directly by LC-MS/MS in urine or plasma. Peptides identical to endogenous hormones (somatropin, insulin, IGF-1) cannot be distinguished by identity alone; detection uses isoform analysis and indirect biomarker panels.

VALIDATED URINE ASSAYS (WADA-RECOGNIZED)

- **GHRPs** (GHRP-2, GHRP-6, hexarelin, ipamorelin) – intact peptide and metabolites; plasma $t_{1/2}$ 15–30 min; urine detection 24–72 h
- **GHRH analogues** (sermorelin, CJC-1295 no-DAC, tesamorelin) – intact peptide; plasma $t_{1/2}$ 7–30 min, urine 1–3 d. **CJC-1295 with DAC** has plasma $t_{1/2}$ 5–8 d (urine 2–3 weeks)
- **BPC-157** – validated urine metabolite assay (2024); plasma $t_{1/2}$ ~30 min; urine 24–48 h
- **TB-500 / thymosin β 4** – urine peptide and metabolites; plasma $t_{1/2}$ 1–3 d; urine 5–7 d
- **Insulin analogues** – analogue-specific signatures (B30 modifications, glargine A21G) by LC-MS/MS in plasma

INDIRECT BIOMARKER PANELS (WHEN DIRECT DETECTION FAILS)

- **Recombinant GH (somatropin)** – identical to natural 22-kDa hGH; direct detection fails. WADA uses an isoform ratio (rhGH suppresses 20-kDa GH) and the GH-2000 score (IGF-1 + P-III-NP); detection windows ~24–48 h (isoform) and up to ~2 weeks (biomarker)
- **IGF-1 / IGF-1 LR3** – short bound half-life (12–15 h); LR3 at low doses is hard to catch; integrated into the GH-2000 score
- **EPO and ESAs** – IEF/SDS-PAGE band pattern plus the Athlete Biological Passport (reticulocyte %, OFF-score)

WHAT CANNOT BE RELIABLY TESTED · MATRIX & STABILITY

- Endogenous-identical biologics outside the somatropin / insulin / IGF-1 indirect-detection framework; niche peptides without validated metabolite signatures (DSIP, dihexa, several Khavinson bioregulators, PE22-28) – no anti-doping assay exists
- **Urine** is the primary WADA matrix (non-invasive, long window); **plasma / DBS** is preferred for short- $t_{1/2}$ peptides, biomarker panels, and the somatropin isoform assay. Urine peptides degrade rapidly at room temperature – cold-chain transport and –20°C/–80°C storage required

Adverse effects & overdose

Class-based adverse-effect profiles, GH-excess recognition and management, and overdose workup for compounded and gray-market peptide use.

GENERAL RISKS (ALL UNAUTHORIZED PEPTIDES)

- **Injection-site infection** from non-sterile preparation or technique.
- **Endotoxin contamination** — 8% of tested samples in one analysis (fever, chills, septic-shock risk).
- **Hypersensitivity and GI effects**, plus contamination with lead or wrong active ingredient — falsified products have carried arsenic up to 10× ICH limits.
- **Skin hyperpigmentation and new or darkening moles** — from melanocyte overstimulation (melanocortin-class peptides) or as a sign of heavy-metal toxicity; chronic arsenic exposure classically causes diffuse hyperpigmentation and skin lesions. New or changing pigmented lesions warrant dermatologic evaluation.

GH SECRETAGOGUES (CJC-1295, IPAMORELIN, TESAMORELIN)

- **Fluid retention** — edema, arthralgia, carpal-tunnel syndrome.
- **Glucose intolerance and elevated IGF-1**, with theoretical malignancy risk from sustained GH/IGF-1 (detailed stats and monitoring in GH-excess section).
- Injection-site reactions, paresthesia, myalgia.

IGF-1 / IGF-1 LR3

- **Hypoglycemia** — LR3 is 2–3× more potent than native IGF-1 acutely, with 4–8× greater cumulative suppression over 4 h; IGF-1 also suppresses glucagon, impairing recovery.
- **Intracranial hypertension, slipped capital femoral epiphysis, scoliosis progression** (esp. adolescents); lymphoid-tissue hypertrophy.

GHK-CU (COPPER PEPTIDE)

- Low cytotoxicity preclinically; **at overdose, copper toxicity dominates** — nausea, vomiting, acute renal failure, hemolytic anemia, hepatic injury. FDA-approved copper histidinate (Zycubo) carries renal-tubular, liver, and hematologic warnings.
- **Hypothetical zinc deficiency** with chronic or high-dose use — copper and zinc compete for intestinal absorption, so sustained copper loading could in theory lower zinc status. Not documented at cosmetic topical doses; consider monitoring copper and zinc with prolonged systemic use.

BPC-157

- Good preclinical safety (LD₁ not reached in mice, rats, rabbits, dogs); **no human safety data**. Injection-site infection and compartment syndrome (requiring fasciotomy) reported after cocktail injection.

GH excess: recognition & workup

Recognizing and diagnosing growth-hormone excess from secretagogue and GHRH-analogue use.

Sustained GH/IGF-1 activation from off-label GHRH analogues (sermorelin, CJC-1295, tesamorelin) and secretagogues (ipamorelin, hexarelin, MK-677) can produce a partial, biochemical, or — rarely — clinically apparent acromegaloid phenotype carrying cardiovascular, metabolic, and oncologic risk. Many patients present after months to years of compounded use without ever having had IGF-1 measured.

CLINICAL SPECTRUM

- **Functional excess** (most common) — fluid retention, edema, arthralgia, carpal-tunnel symptoms, paresthesia, sleep disturbance; largely reversible off-drug.
- **Acromegaloid features** (prolonged/supraphysiologic use) — soft-tissue swelling of hands and feet (ring or shoe tightening is often the first clue), facial coarsening, prognathism, macroglossia, skin-tag proliferation, hyperhidrosis; may be irreversible.
- **Metabolic** — impaired glucose tolerance, new-onset T2DM (tesamorelin HR 3.3 vs placebo), insulin resistance, dyslipidemia.
- **Cardiovascular** — hypertension, LV hypertrophy, diastolic dysfunction, OSA.
- **Oncologic concern** — sustained IGF-1 theoretically promotes colorectal, breast, and prostate proliferation; no causal data in secretagogue users, but the association in endogenous acromegaly is well-established.

WHEN TO SUSPECT & HOW TO WORK UP

Suspect in any GH-axis-peptide user with new arthralgia, paresthesia, ring-size change, fluid retention, new T2DM/OSA/hypertension, asymmetric soft-tissue change (hands/feet/jaw), or an IGF-1 above the age-adjusted ULN on monitoring.

- **Serum IGF-1 (age/sex-adjusted)** — the workhorse; single random draw, unaffected by pulsatile GH; repeat in 2–4 wks if equivocal.
- **OGTT with GH suppression** — 75 g, GH at 0–120 min; normal nadir $< 1 \mu\text{g/L}$ (< 0.4 on ultrasensitive assays). Confirms excess but doesn't distinguish exogenous from endogenous.
- **Pituitary MRI** — if IGF-1 stays elevated > 8 wks after stopping, or biochemistry is severe, to exclude a somatotroph adenoma.
- **Ancillary** — fasting glucose/HbA1c, lipids, ECG, echo if cardiac symptoms, colonoscopy if age > 50 , sleep study if OSA features.

GH excess management & overdose

IGF-1 targets, dose reduction, and a structured workup for compounded-peptide overdose.

IGF-1 targets. On therapy: upper-half of the age-adjusted range, never above ULN.

Above ULN — reduce dose or lengthen interval; recheck in 4–6 wks.

> 1.5× ULN — discontinue; evaluate for acromegaloïd features; consider OGTT-GH and pituitary MRI.

Tesamorelin — 47% of treated patients exceeded ULN at 26 wks in the pivotal trial; monitor at baseline, 4–6 wks, then every 3 months.

MANAGEMENT

- **Discontinue** the agent — functional features reverse in 8–12 wks; structural acromegaloïd changes may not.
- **Address comorbidities** — hypertension, glucose intolerance, OSA; cardiology referral if LVH or symptoms.
- **If IGF-1 stays elevated > 8–12 wks off therapy** → full workup for endogenous GH excess (coincident acromegaly has been unmasked by secretagogue use).
- **Counsel against resumption**; document informed consent if the patient continues.

OVERDOSE ASSESSMENT

- **Stabilize & characterize:** ABCs first (EMS for instability); intent (accidental vs self-harm); substance, formulation, and source; intended vs actual dose and units (mg/mcg/"units"); time course and stacking; patient factors; current symptoms by system.
- **No specific antidote exists** — management is symptomatic and supportive.
- **Workup, symptom-guided:** electrolytes, creatinine, ALT/AST; CK if muscle pain or dark urine; glucose for IGF-1/GH peptides; **C-peptide** to separate exogenous insulin (suppressed) from endogenous hyperinsulinism (elevated); troponin/ECG for chest pain; rule out co-ingestions (acetaminophen, salicylate, ethanol).
- **Pearls:** "any dose is an overdose" for most of these; always ask about injectable peptides (patients won't volunteer); prolonged monitoring for long-acting agents (CJC-1295 t½ 5.8–8.1 d); adolescent use risks premature epiphyseal closure. Report reactions to Canada Vigilance / FDA MedWatch.

The regulatory landscape

Route, claim, and the FDA hierarchy: how the same peptide can be legal one way and illegal another.

Route matters more than molecule. Across most jurisdictions, route of administration — not molecular identity — drives regulatory status. Injectable peptide formulations face near-universal prohibition or severe restriction; oral and topical formulations have recognized legal pathways in several countries.

FDA-approved drugs have completed Phase 3 and carry agency-reviewed indications, dosing, and safety language — most of the metabolic, bone, somatostatin, and reproductive entries here. Note that *orphan-drug designation is not marketing approval* (thymosin α 1 holds three orphan designations but no NDA — a distinction often conflated in marketing).

503A vs 503B compounding. 503A pharmacies prepare patient-specific prescriptions; 503B facilities manufacture cGMP batches. Both require a COA documenting identity, purity ($\geq 98\%$ HPLC), sterility, and endotoxin, and the bulk substance must be on FDA Category 1, in a USP/NF monograph, or in an approved product.

The shifting 503A list. A Sept 2023 FDA Category 2 action restricted 17 commonly compounded peptides; in Sept 2024 the FDA removed AOD-9604, CJC-1295, ipamorelin, thymosin α 1, and Selank pending PCAC review. The next decisive milestone is the **July 23–24, 2026 PCAC meeting** (Docket FDA-2025-N-6895), reviewing BPC-157, KPV, TB-500, MOTS-c, DSIP, Semax, and Epitalon for possible 503A inclusion. Outcomes are unknown, and the narrow indications under review may not match the broader off-label uses driving demand.

Current FDA 503A Category 2, compounding restricted: BPC-157 · TB-500 (thymosin β 4) · melanotan II · KPV · semax · hexarelin · GHRP-2 · GHRP-6 · MK-677 (ibutamoren) · epitalon · MOTS-c · LL-37 · DSIP · dihexa · PEG-MGF · GHK-Cu (injectable only)

GHK-Cu is the exception — Category 1 for non-injectable routes (topical/oral) and Category 2 for injectable, making it the only peptide here with a recognized US compounding pathway. **Health Canada** recognizes BPC-157 and GHK-Cu as approved chemical isolates for oral/topical use only.

Jurisdictional snapshot. *Australia (TGA):* BPC-157 Schedule 4 + Appendix D — possession without authority is a criminal offence. *UK (MHRA):* not licensed for human use. *EU:* no EMA authorization; member-state enforcement. *New Zealand:* prescription medicine since 2023. *UAE/GCC:* Federal Decree-Law 38 (2024) permits licensed compounding under valid prescriptions to GCP/USP 795/797. *WADA:* Section S0 prohibits all non-approved substances; no TUE.

Research-use-only (RUO) products carry most gray-market trade. The "not for human consumption" disclaimer offers *no legal protection* — the FDA regulates by intended use, inferred from marketing, dosing calculators, and reviews (50+ warning letters to RUO vendors in Sept 2025 alone). Independent assays have found 60–110% of stated content plus endotoxin, residual TFA, and heavy-metal contamination.

TWO LEGAL PITFALLS FOR PROVIDERS

"Off-label" doesn't apply to unapproved drugs. Off-label use is a legal pathway only for drugs holding at least one FDA-approved indication. Prescribing BPC-157 or any never-approved peptide is use of an *unapproved new drug* — IND, Expanded Access, and Right-to-Try do not cover routine wellness use. **And "recommending" is not a liability shield:** directing patients to source research-grade peptides for self-administration is practicing medicine; liability and board jurisdiction attach regardless of whether the practitioner handled the product, and most malpractice policies exclude unapproved-drug claims. **Bottom line:** if a peptide is not FDA-approved, you are operating outside the drug-approval framework — document the rationale, product source, and informed consent.

Cosmetic, compounded, or supplement?

Three regulatory regimes for the same molecule

The *same* peptide can be sold under three very different regulatory regimes depending on its **route and the claims made** — not its molecular identity. The regime, not the molecule, determines what is lawful.

COSMETIC — TOPICAL, APPEARANCE CLAIMS ONLY

Under the FD&C Act and the Modernization of Cosmetics Regulation Act (MoCRA, 2022), a topical product that makes only *appearance* claims (moisturizing, improving the look of fine lines) is a cosmetic: no premarket approval, but safety and labeling rules apply. A *therapeutic* claim (healing, regenerating tissue, treating a condition) reclassifies it as an unapproved drug.

Example: a GHK-copper cream sold for skin appearance is a cosmetic; the identical cream marketed to "regenerate tissue" crosses into drug territory.

COMPOUNDED — PRESCRIPTION, 503A / 503B

A licensed pharmacy may compound a peptide for an individual patient prescription (503A) or as cGMP batches for clinics (503B), but only when the bulk substance is eligible — on FDA Category 1, in a USP/NF monograph, or in an approved product — with a COA for identity, purity, and sterility. **Example: GHK-copper is Category 1 for non-injectable compounding**; most injectable research peptides are not eligible and cannot be lawfully compounded.

DIETARY SUPPLEMENT — ORAL, DSHEA

Marketing a peptide as an oral "dietary supplement" under DSHEA is the most legally contested route. Most synthetic peptides do not meet the statutory definition of a dietary ingredient and lack a New Dietary Ingredient (NDI) notification; the FDA has stated that several — **including BPC-157** — are excluded from the supplement category and treated as unapproved drugs.

Example: oral BPC-157 capsules are widely sold as "supplements," but the FDA has issued warning letters rejecting that classification.

Bottom line: route plus claim define the regime — the same peptide can be a lawful cosmetic, a prescription compound, or an unlawful "supplement" depending on how it is sold.

Case study: Lilly v. FDA

Retatrutide and the >40-amino-acid boundary (2023–2025)

THE MOLECULE AND THE DISPUTE

Retatrutide (Lilly's investigational GLP-1/GIP/glucagon triple agonist) has a **39-alpha-amino-acid** backbone plus a covalent side chain of one γ -glutamate and one ADO residue — the latter not an alpha amino acid. Counted cumulatively it has 41 residues; counted as alpha amino acids, 39 (sequence patent-disclosed: US20230143781A1). In November 2023 Lilly asked the FDA to classify it as a biological "protein" under the BPCIA's >40-amino-acid threshold, counting all 41 residues. The FDA counted only the 39 backbone alpha amino acids, excluded the ADO residue, and ruled it a drug under the FD&C Act — declining to designate it "analogous to a protein."

THE RULING AND WHY IT MATTERS

Lilly sued (S.D. Ind., 1:24-cv-01503); on **September 30, 2025** the court vacated the FDA's decision and remanded, finding its narrow reading of "analogous to a protein" unreasonable — but it did not finally establish that retatrutide is a biologic. Classification remains procedurally unsettled pending FDA action on remand. As a single district-court ruling (subject to possible appeal), it is persuasive rather than settled national law, but it is the cleanest case to date on how the >40-residue threshold pivots on which residues you count — with large commercial stakes: BLA vs NDA pathways, 12-year biologic vs shorter NCE exclusivity, biosimilar vs generic competition, and very different 503A compounding posture.

CLINICAL RELEVANCE FOR PROVIDERS

For prescribers and compounding pharmacies, classification as a biologic would put generic competition under the BPCIA rather than the FD&C Act — shifting the market-exclusivity and biosimilar timeline — and would change whether and how retatrutide could ever enter a compounding pathway, since 503A compounding of biologics operates under different authority than for small-molecule drugs.

Case citation: Eli Lilly and Co. v. FDA, 1:24-cv-01503, S.D. Ind. (filed Sep. 18, 2024; decided Sep. 30, 2025).

References & key trials

For each peptide, a selected bibliography – pivotal trials, mechanism papers, or recent reviews. Each entry carries a short visual tag identifying the trial, label, or source type, with PMIDs or DOIs given for the primary literature and docket or regulatory citations for legal and approval references. This is a curated, not exhaustive, list – the book's GRADE letters and clinical caveats rest on this evidence base.

METABOLIC · INCRETINS AND INSULIN

Insulin

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At the National Institutes of Health, Hannah-Shmouni led pituitary and adrenal research and directed translational work on cortisol, aldosterone, and IGF-1 dysregulation. He received the Novo Nordisk, Inc. award for growth-hormone research on GH excess in neurofibromatosis 1 syndrome. His work has shaped contemporary understanding of endocrine disease and stress physiology, and is reflected in more than 100 peer-reviewed publications. Today, he serves as Managing Partner at FHS Capital, Assistant Clinical Professor at the University of British Columbia, and Founder and CEO of Hormonal.ai, a clinical AI platform translating evidence-based hormone and longevity science into tools clinicians can use in everyday practice.

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Colophon

About this guide, Hormonal.ai, and Hormonal Press.

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